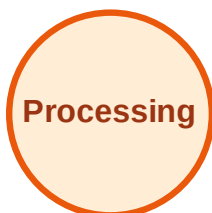


BFS Traversal

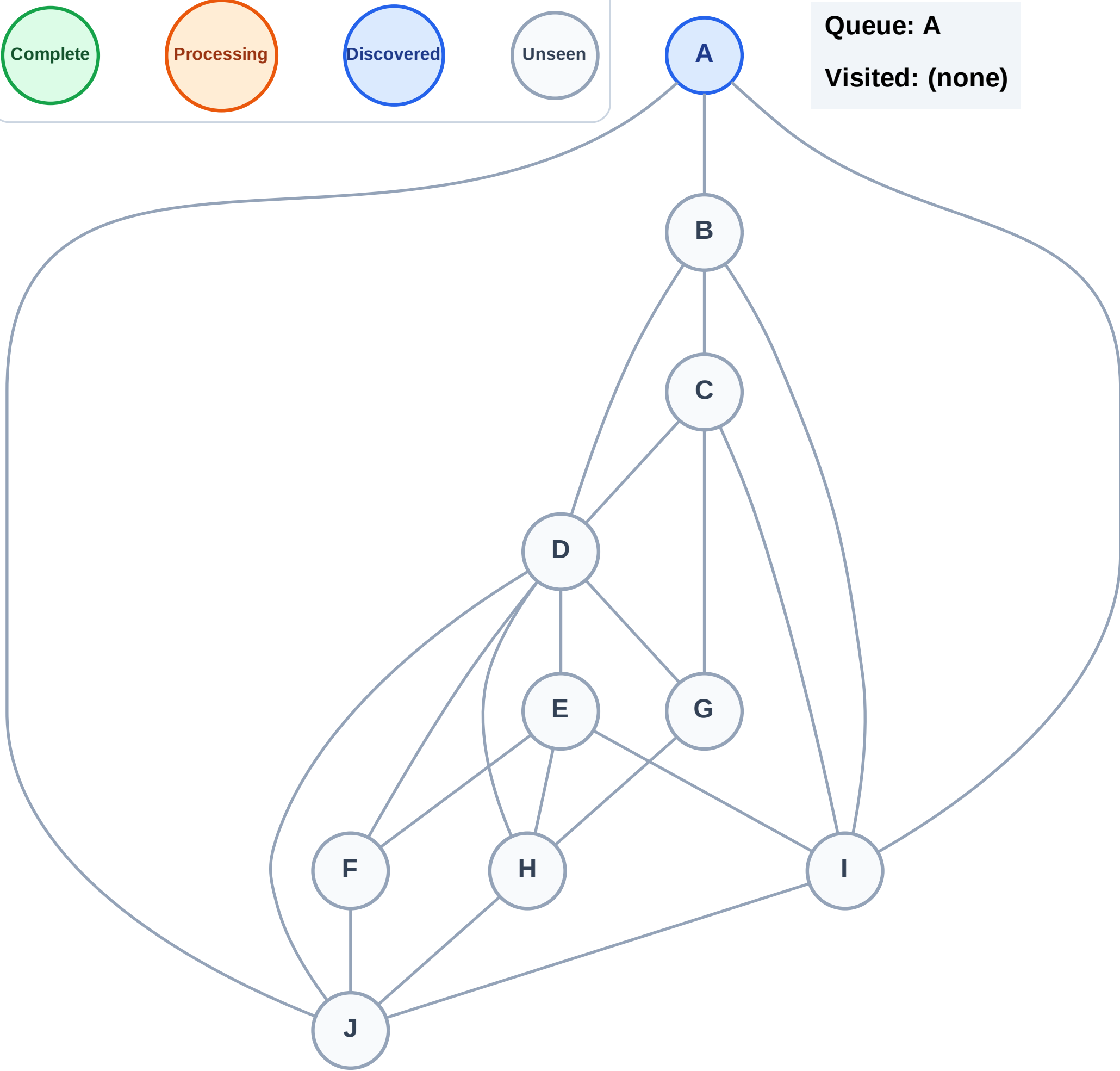
Step 1: Start at A

Legend



Queue: A

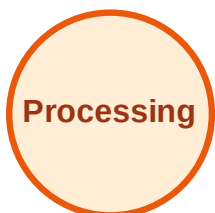
Visited: (none)



BFS Traversal

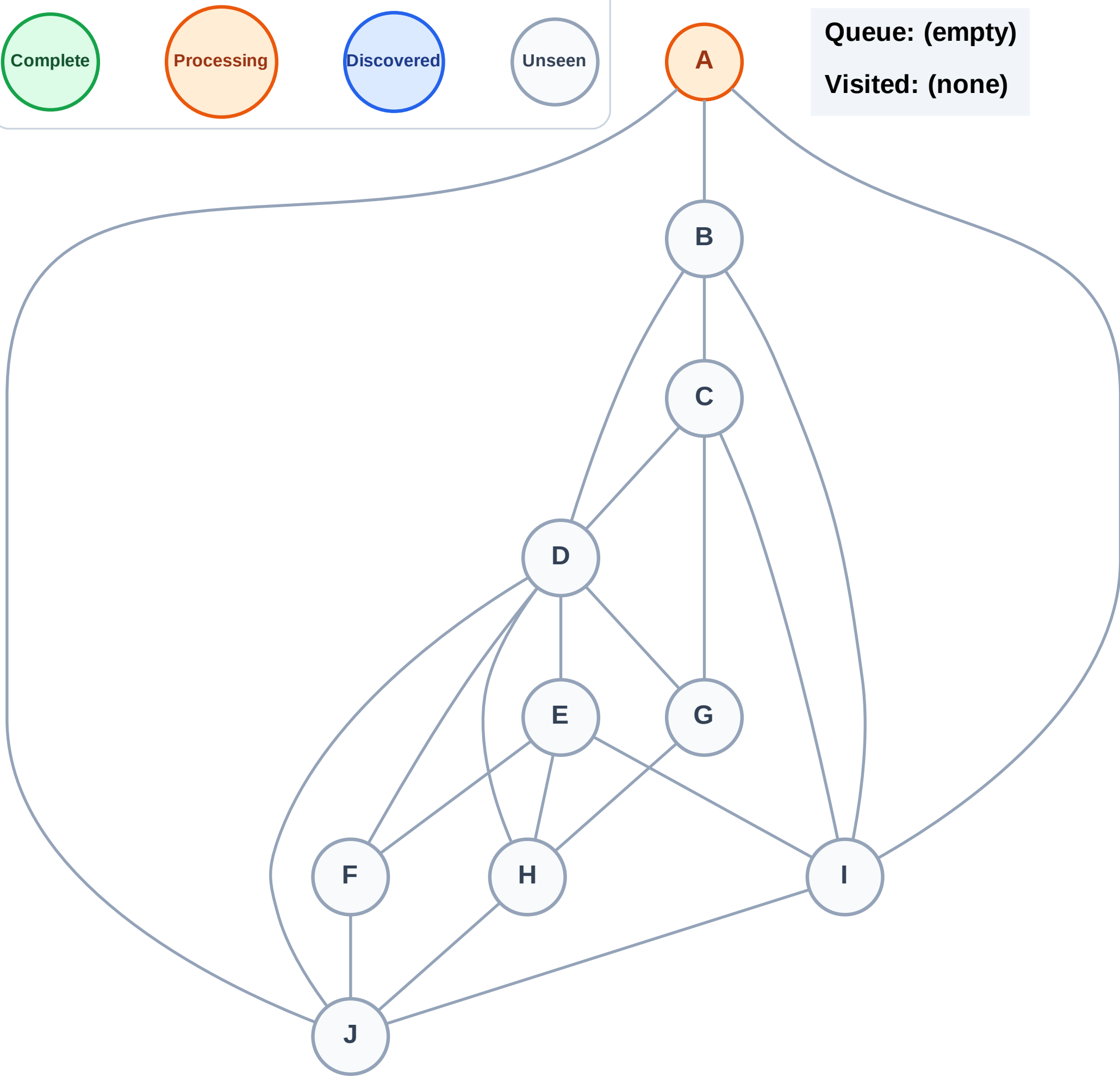
Step 2: Process node A

Legend



Queue: (empty)

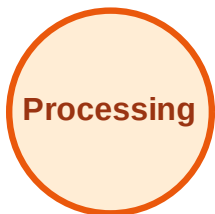
Visited: (none)



BFS Traversal

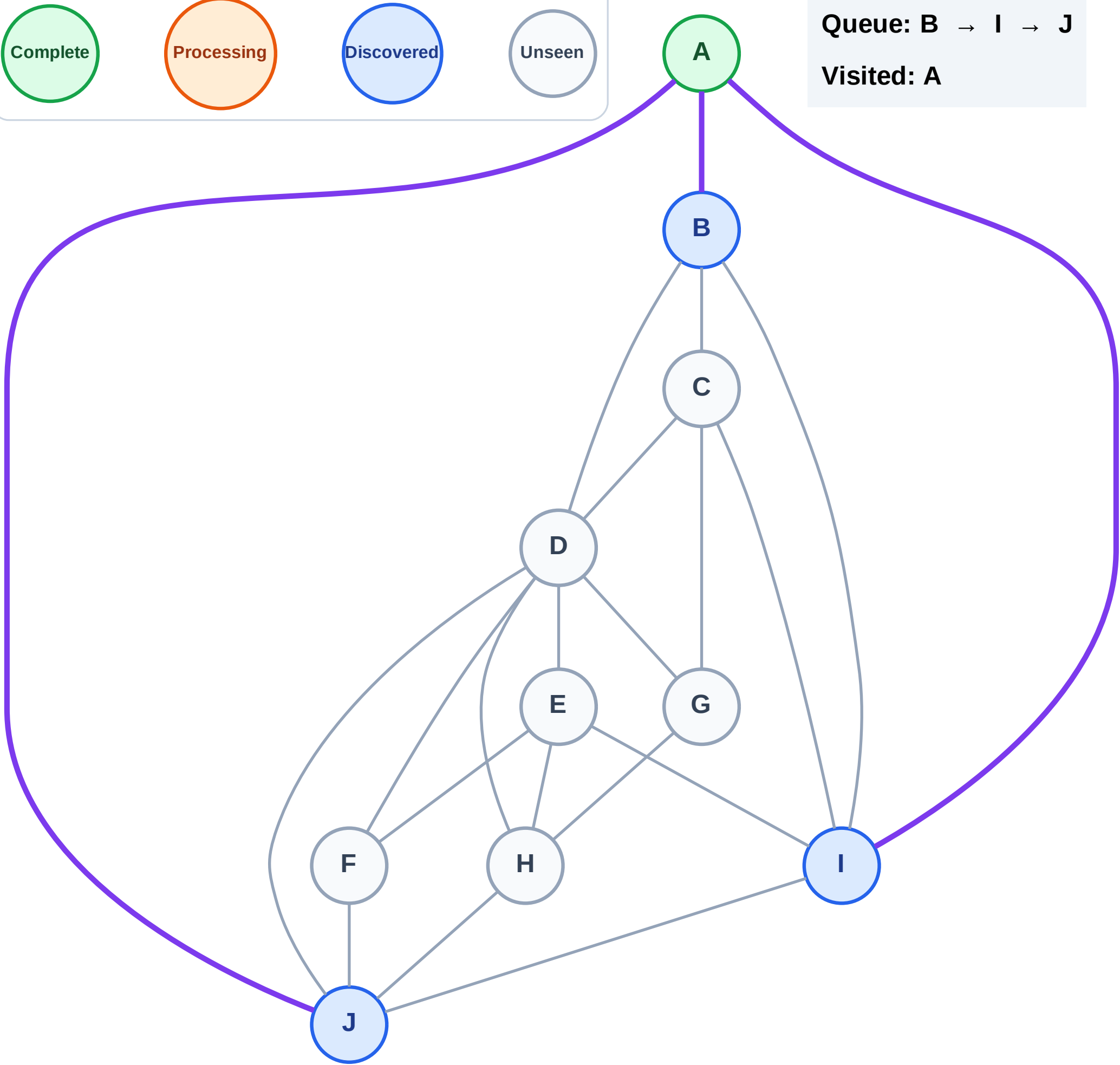
Step 3: Discover B, I, J from A

Legend



Queue: B → I → J

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

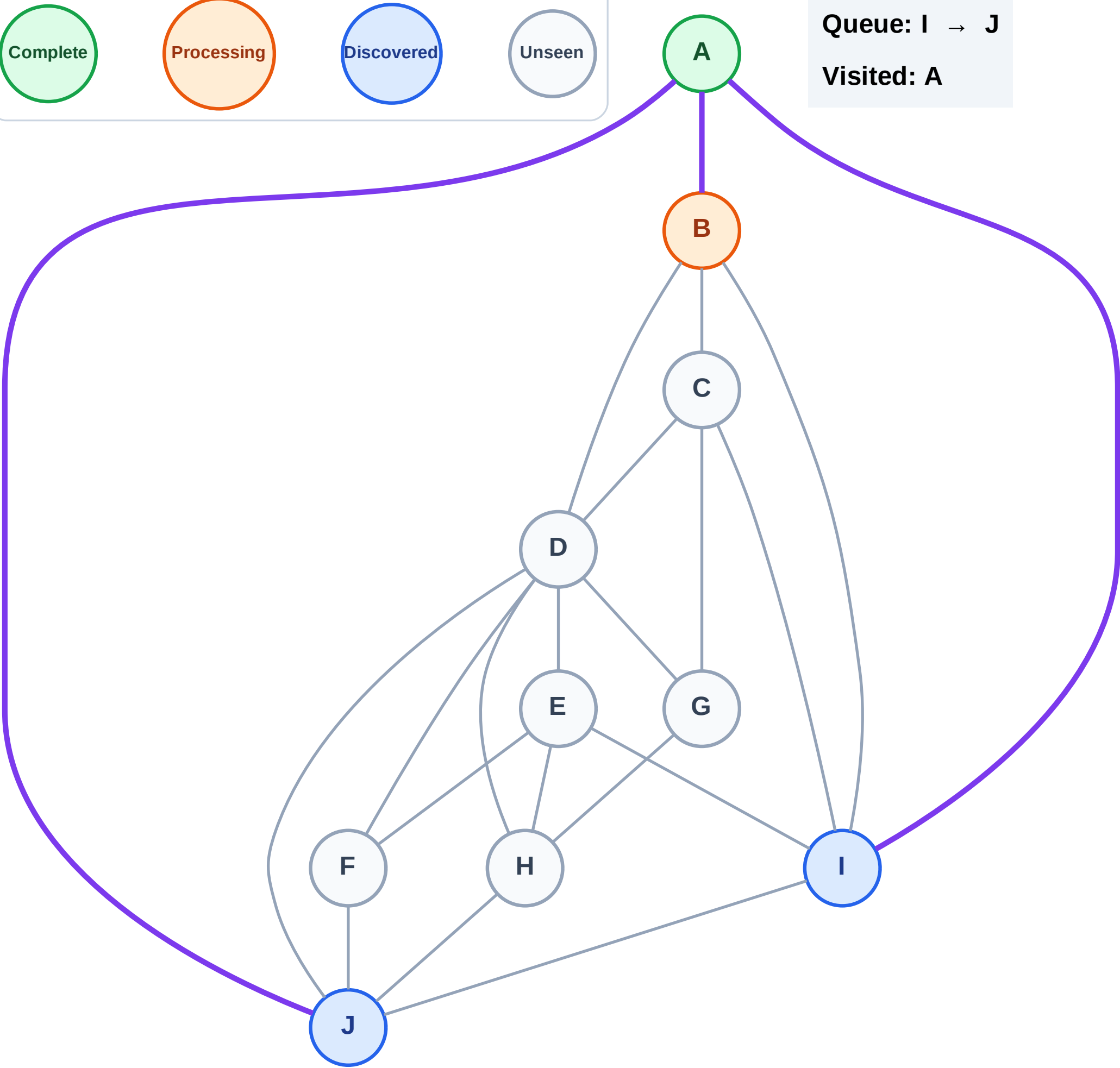
Processing

Discovered

Unseen

Queue: I → J

Visited: A



BFS Traversal

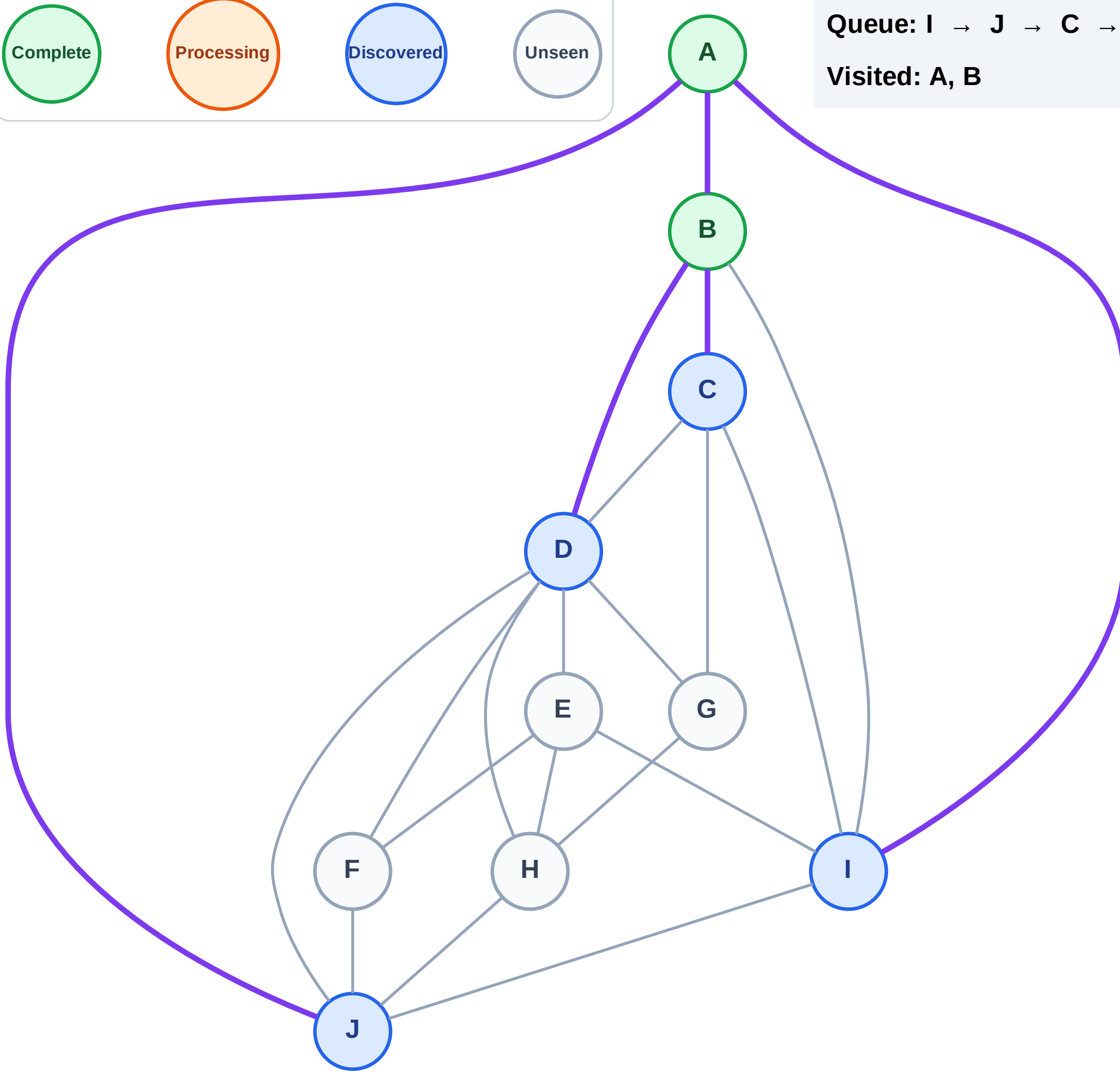
Step 5: Discover C, D from B

Legend



Queue: I → J → C → D

Visited: A, B



BFS Traversal

Step 6: Process node I

Legend

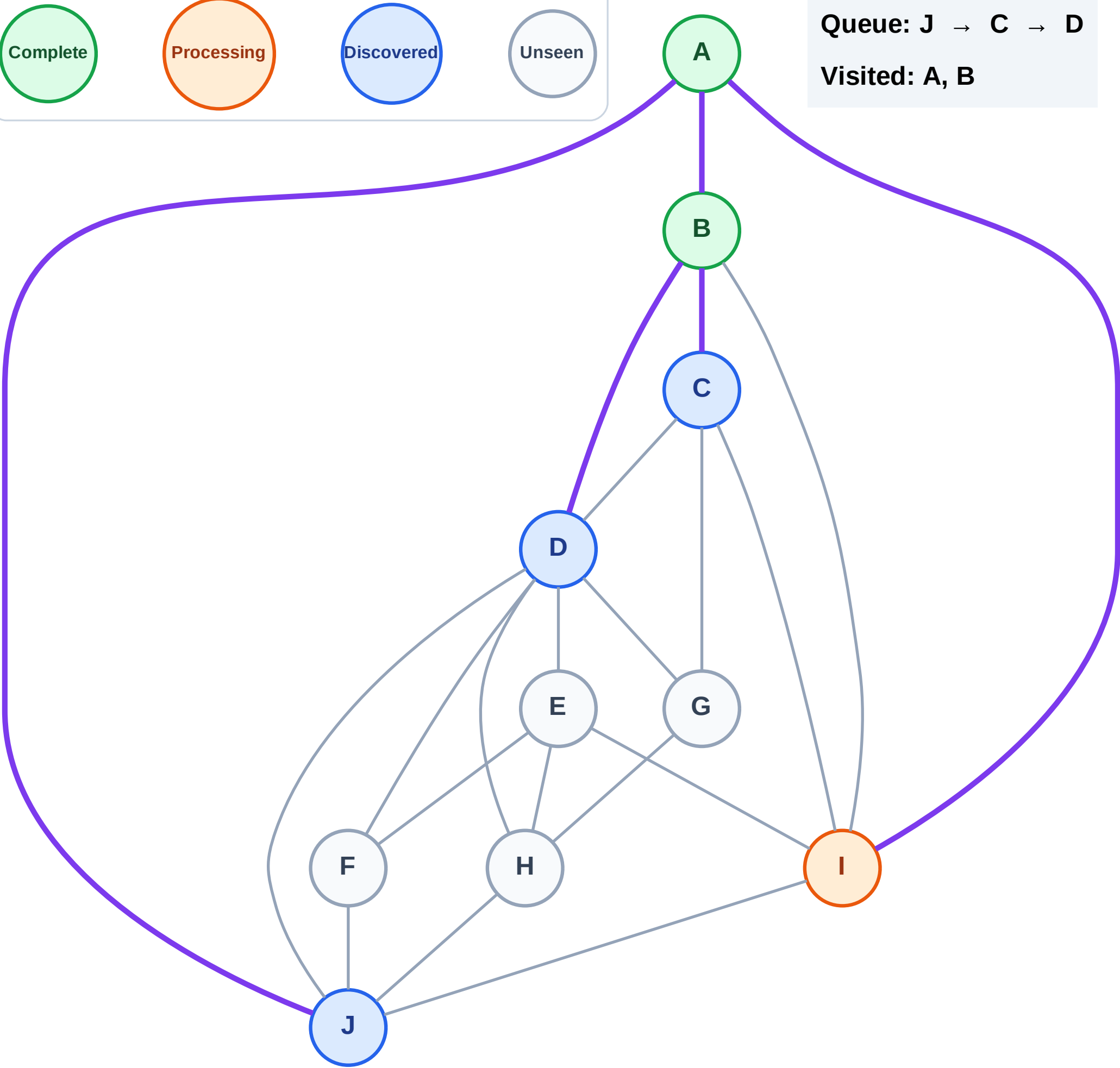
Complete

Processing

Discovered

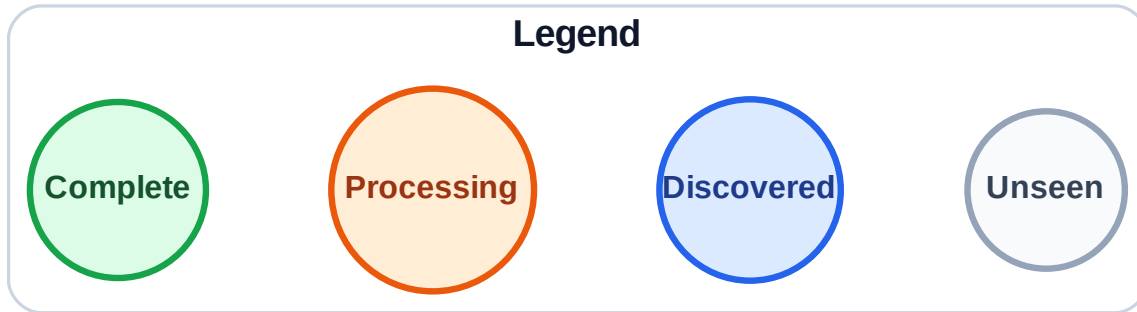
Unseen

Queue: J → C → D
Visited: A, B



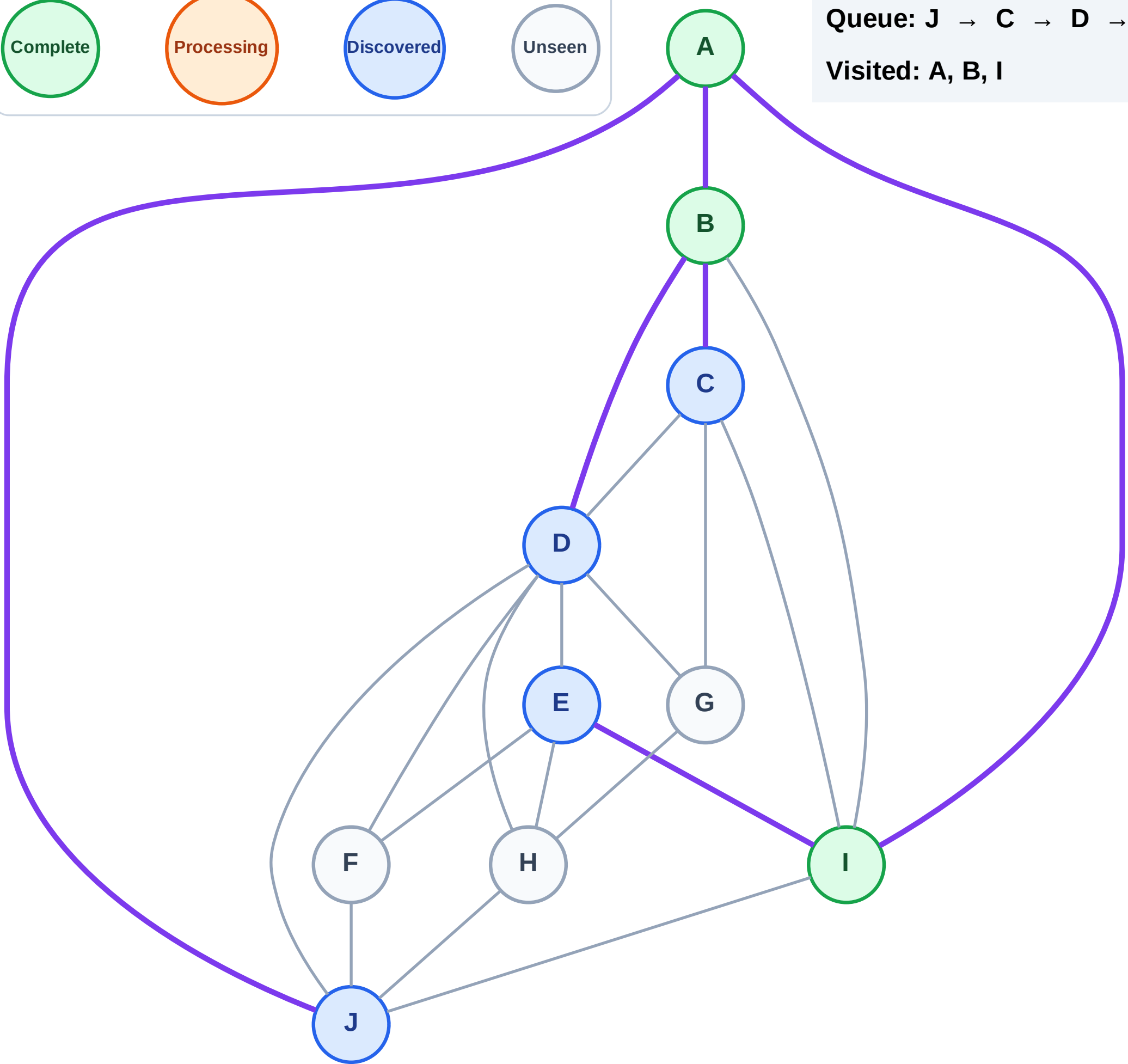
Step 7: Discover E from I

Step 7: Discover E from I



Queue: J → C → D → E

Visited: A, B, I



BFS Traversal

Step 8: Process node J

Legend

Complete

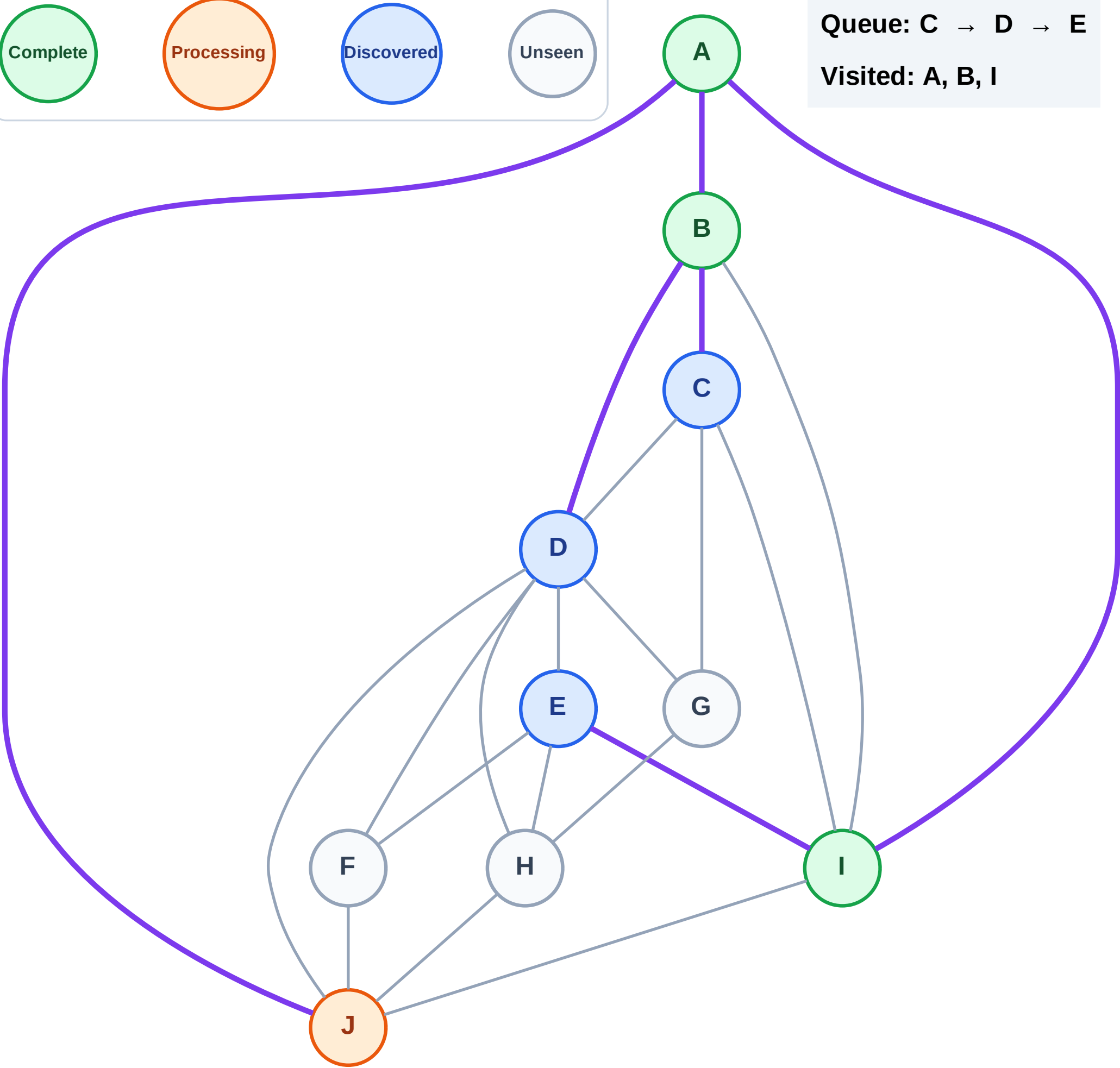
Processing

Discovered

Unseen

Queue: C → D → E

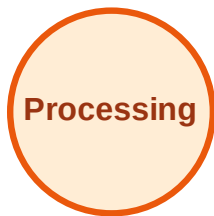
Visited: A, B, I



BFS Traversal

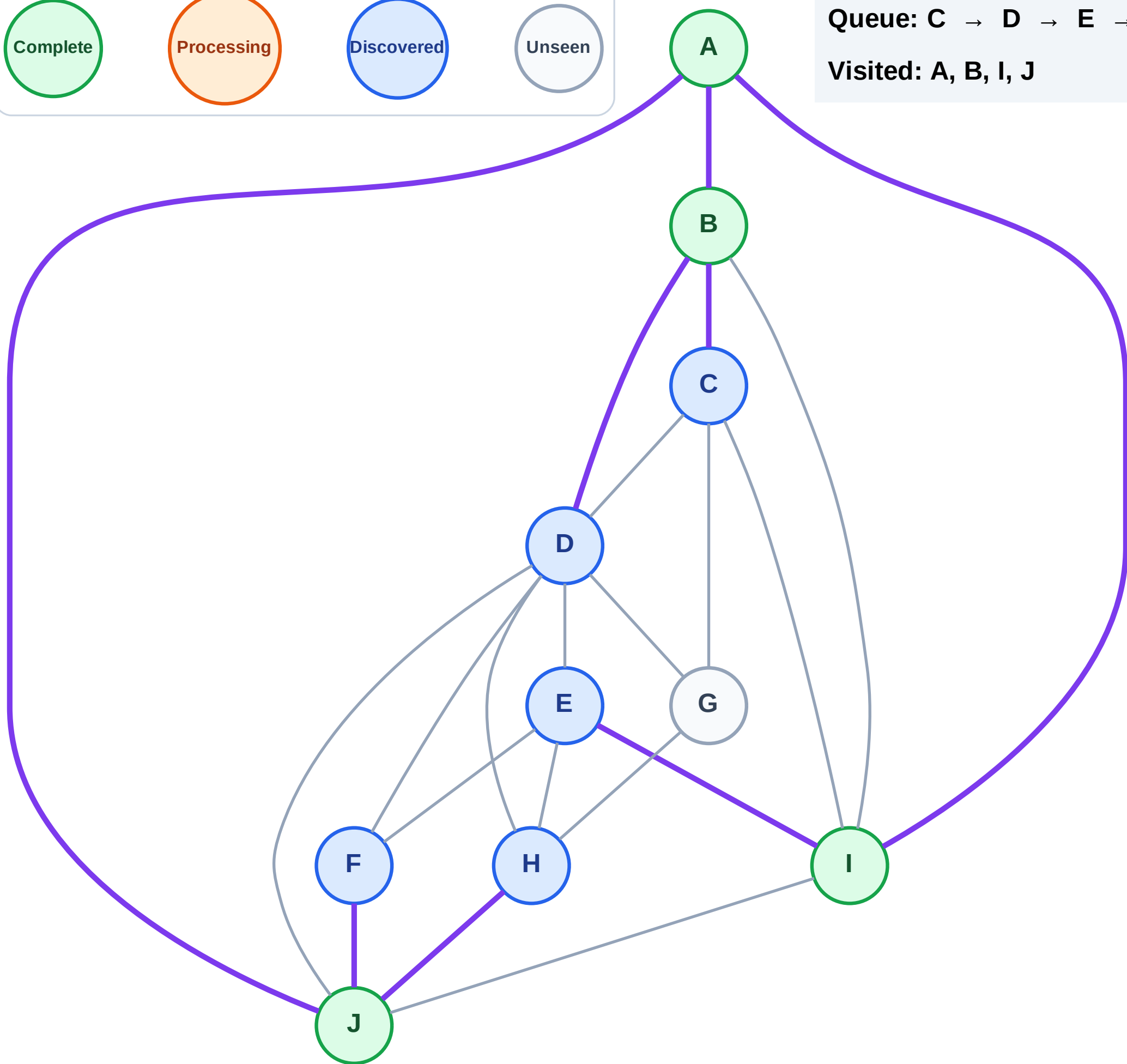
Step 9: Discover F, H from J

Legend



Queue: C → D → E → F → H

Visited: A, B, I, J



BFS Traversal

Step 10: Process node C

Legend

Complete

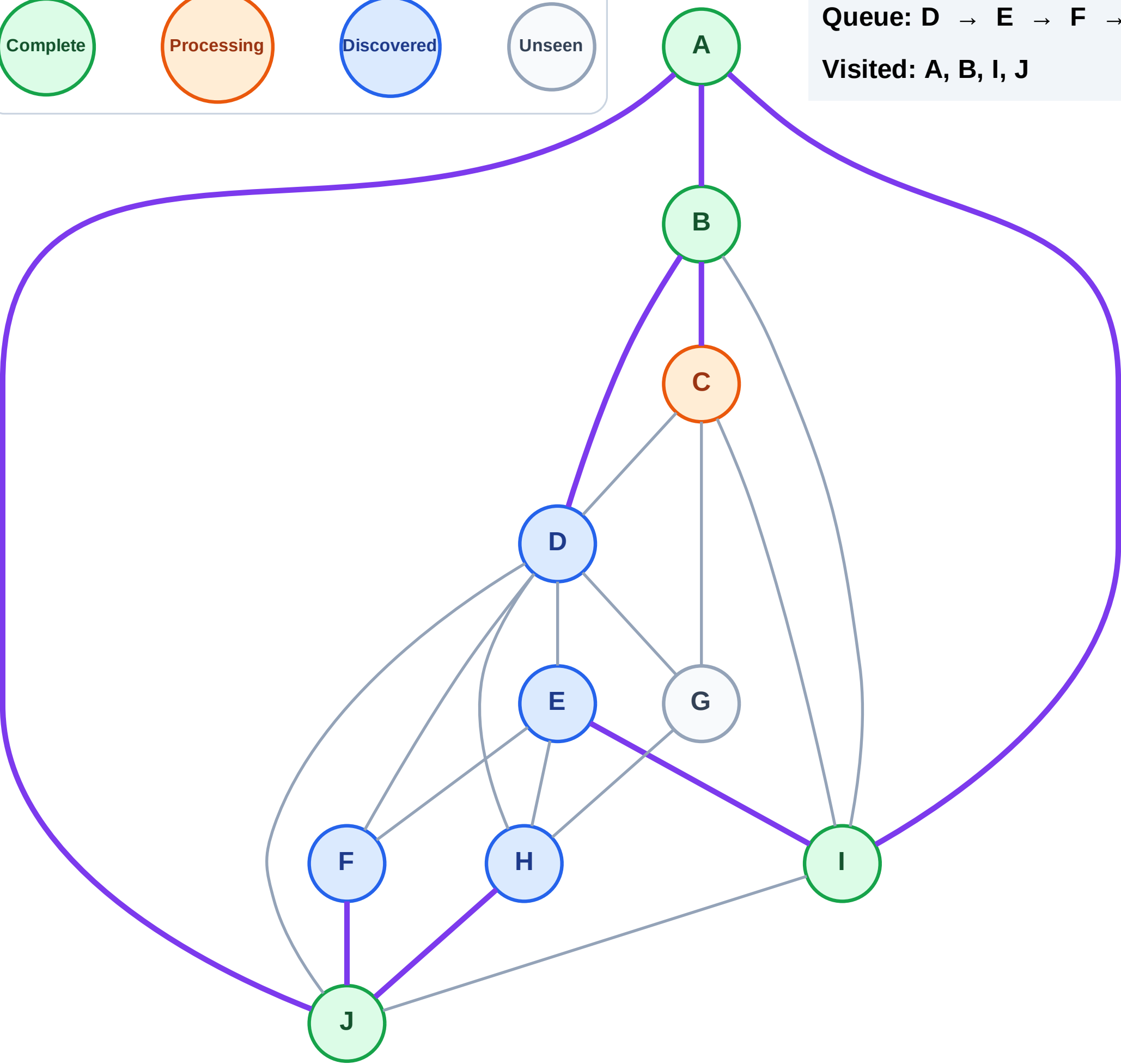
Processing

Discovered

Unseen

Queue: D → E → F → H

Visited: A, B, I, J



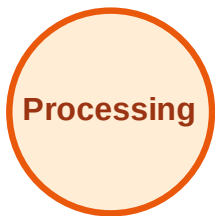
BFS Traversal

Step 11: Discover G from C

Legend



Complete



Processing



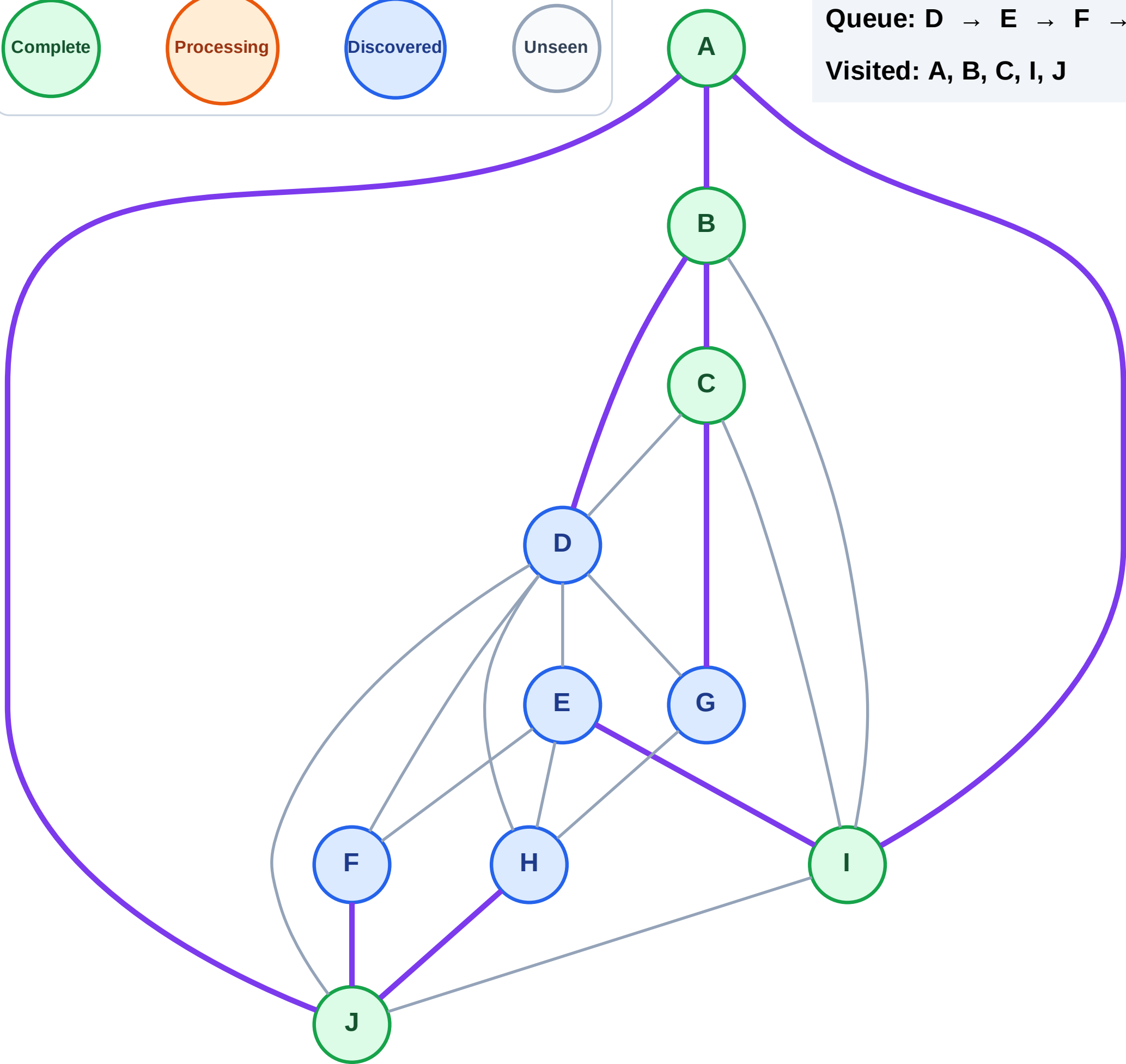
Discovered



Unseen

Queue: D → E → F → H → G

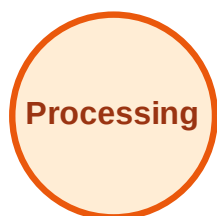
Visited: A, B, C, I, J



BFS Traversal

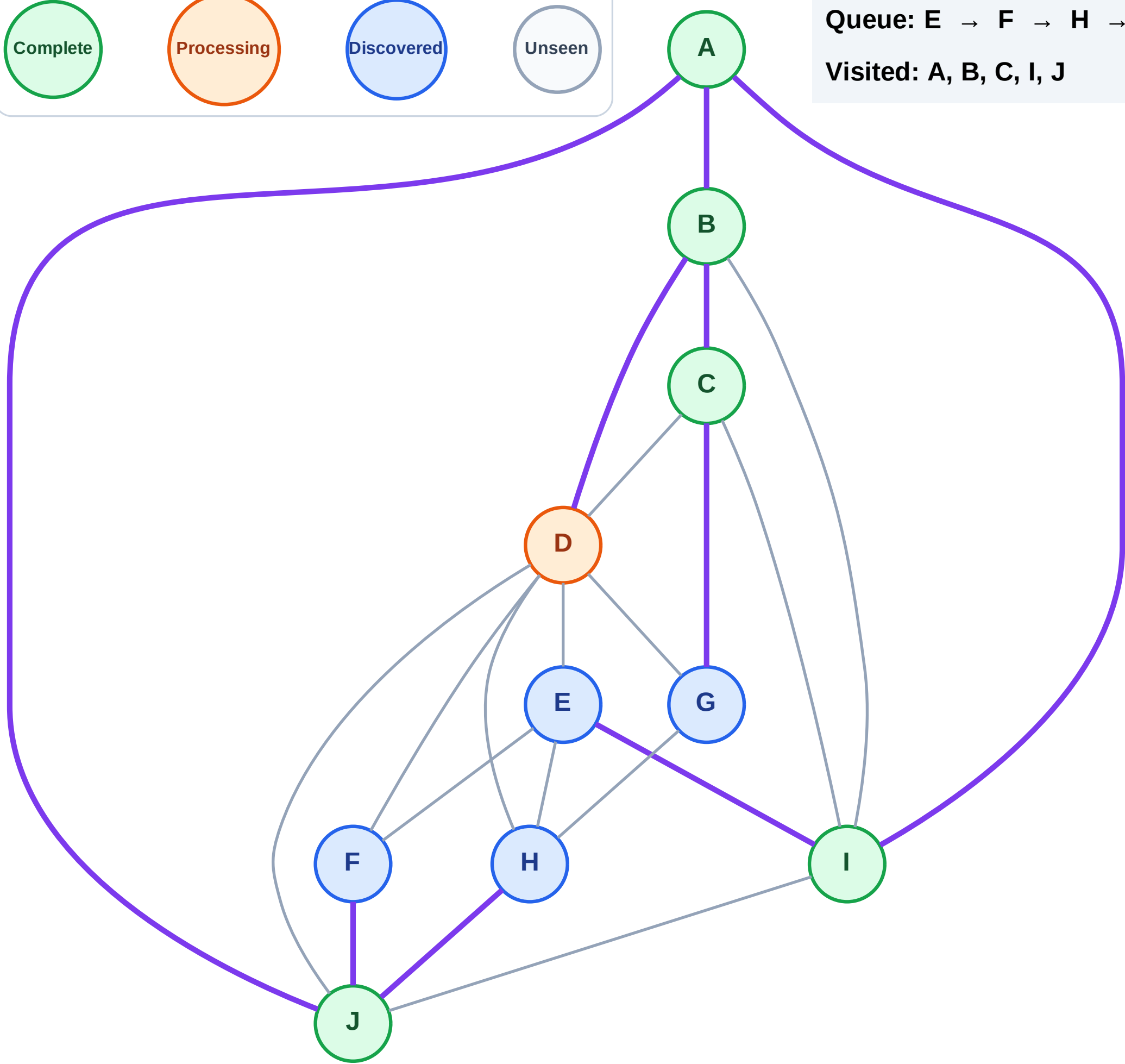
Step 12: Process node D

Legend



Queue: E → F → H → G

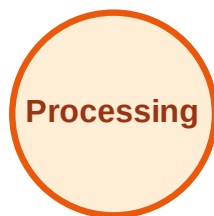
Visited: A, B, C, I, J



BFS Traversal

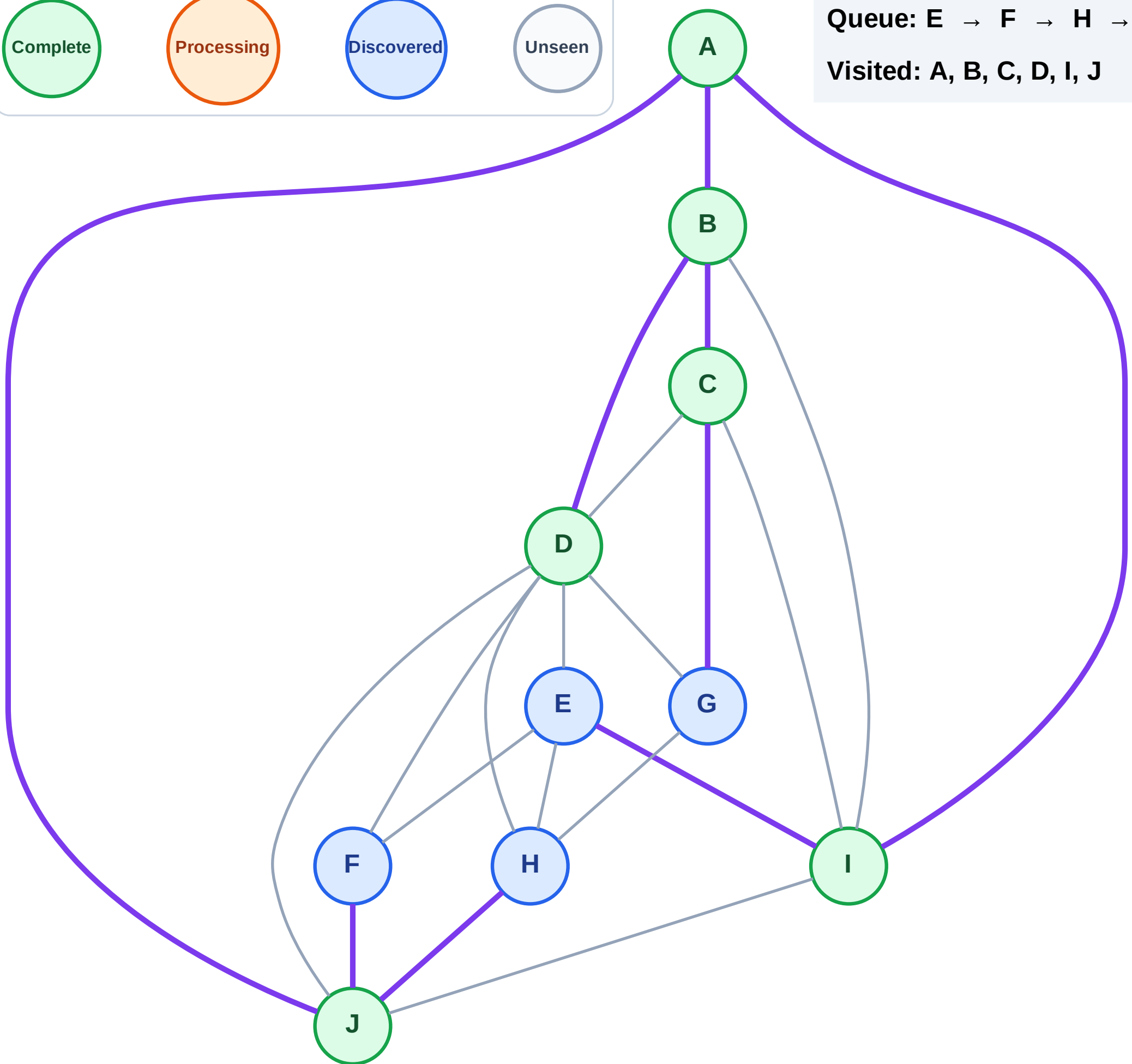
Step 13: No new neighbors from D

Legend



Queue: E → F → H → G

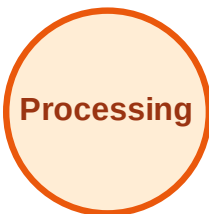
Visited: A, B, C, D, I, J



BFS Traversal

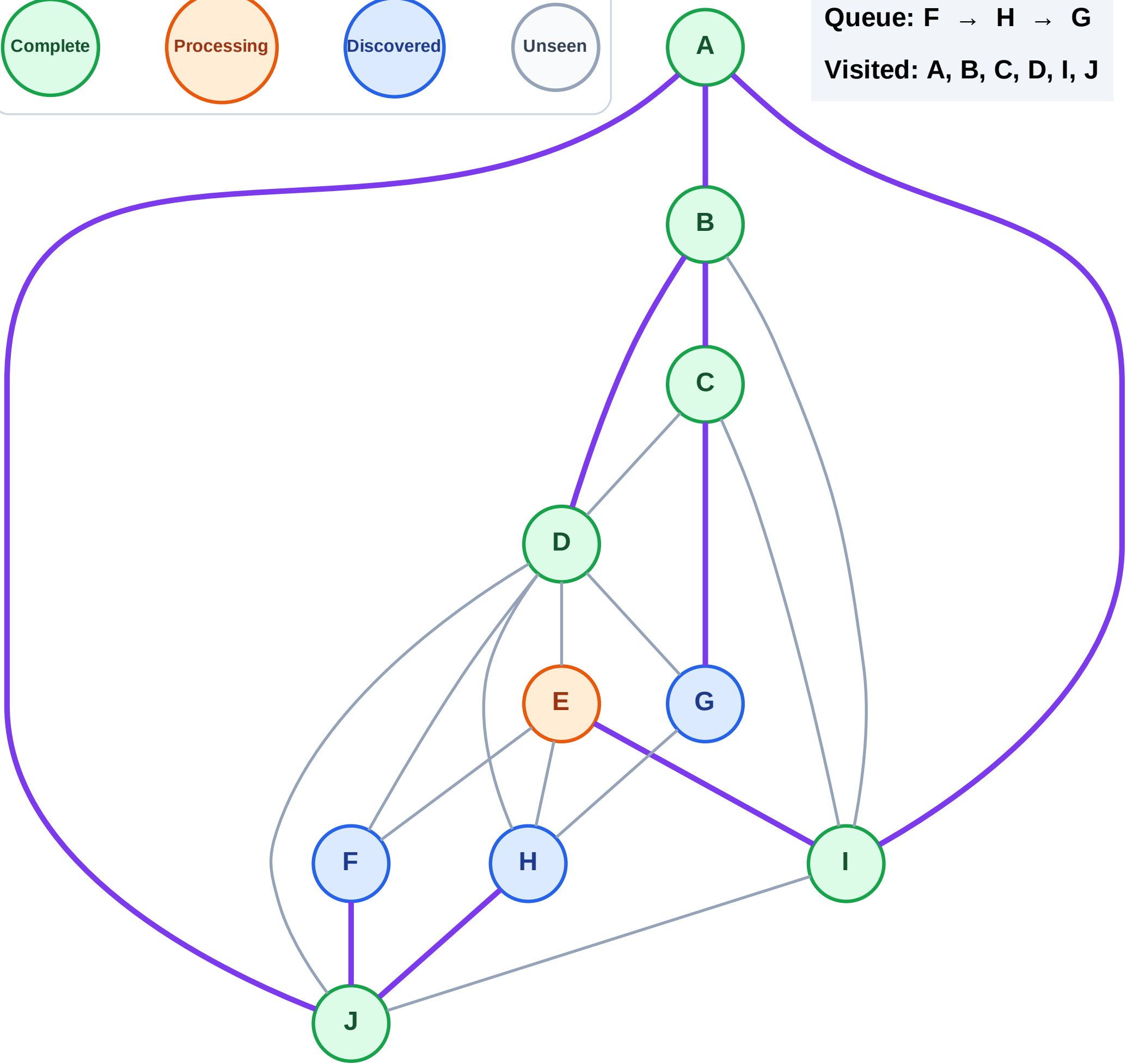
Step 14: Process node E

Legend



Queue: F → H → G

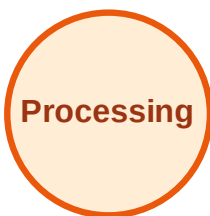
Visited: A, B, C, D, I, J



BFS Traversal

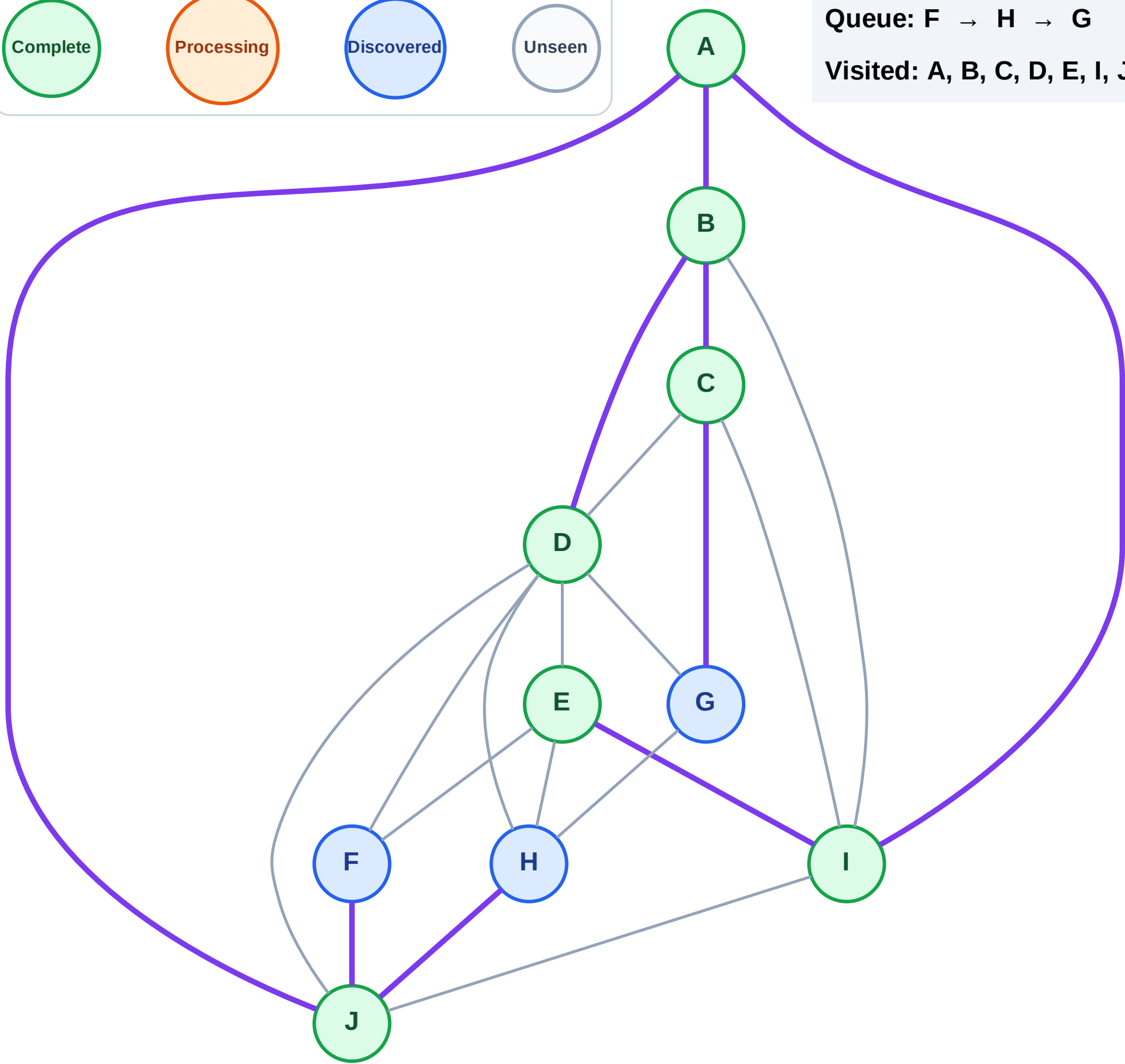
Step 15: No new neighbors from E

Legend



Queue: F → H → G

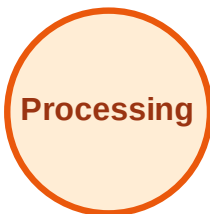
Visited: A, B, C, D, E, I, J



BFS Traversal

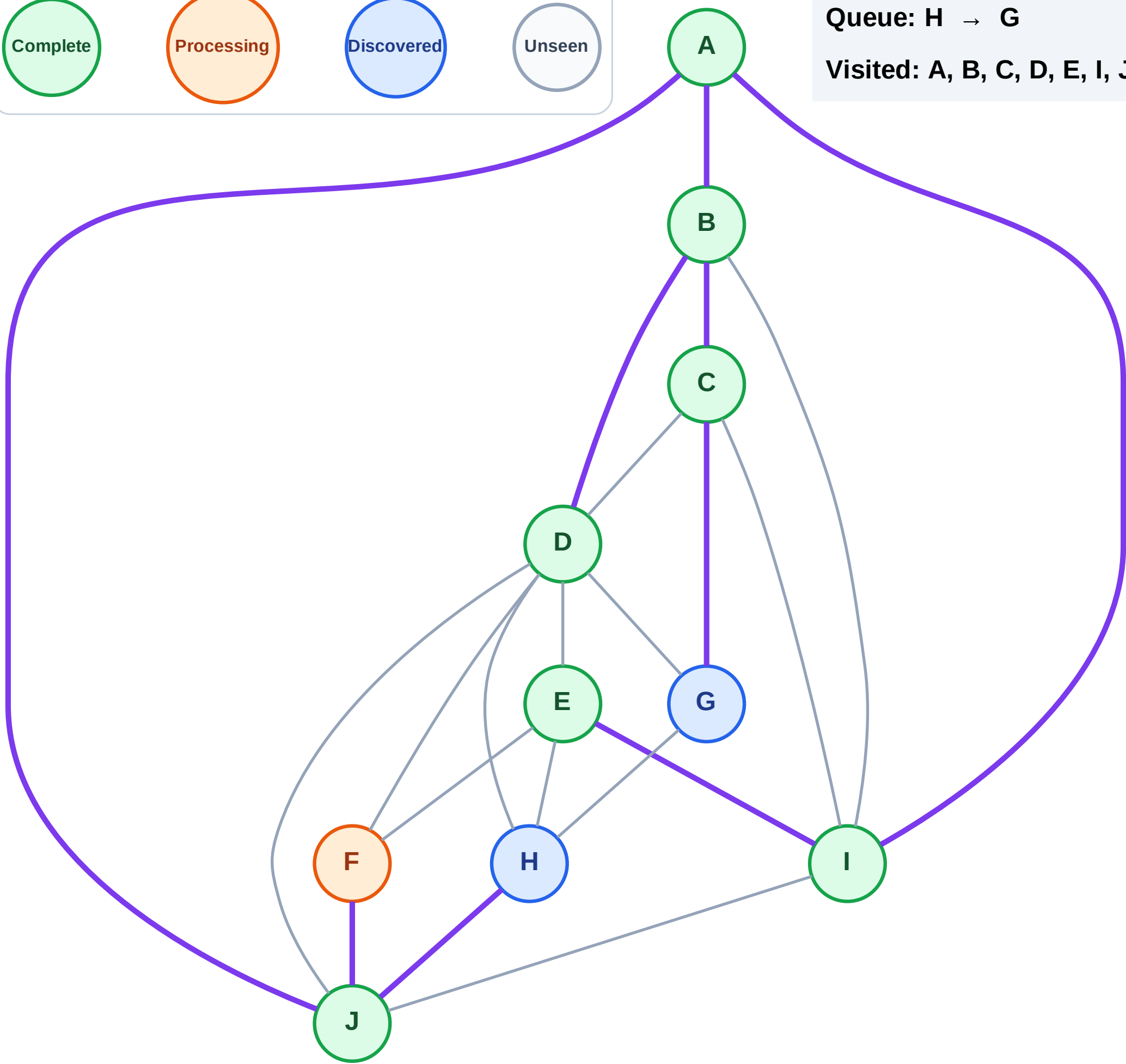
Step 16: Process node F

Legend



Queue: H → G

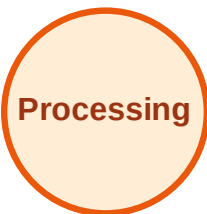
Visited: A, B, C, D, E, I, J



BFS Traversal

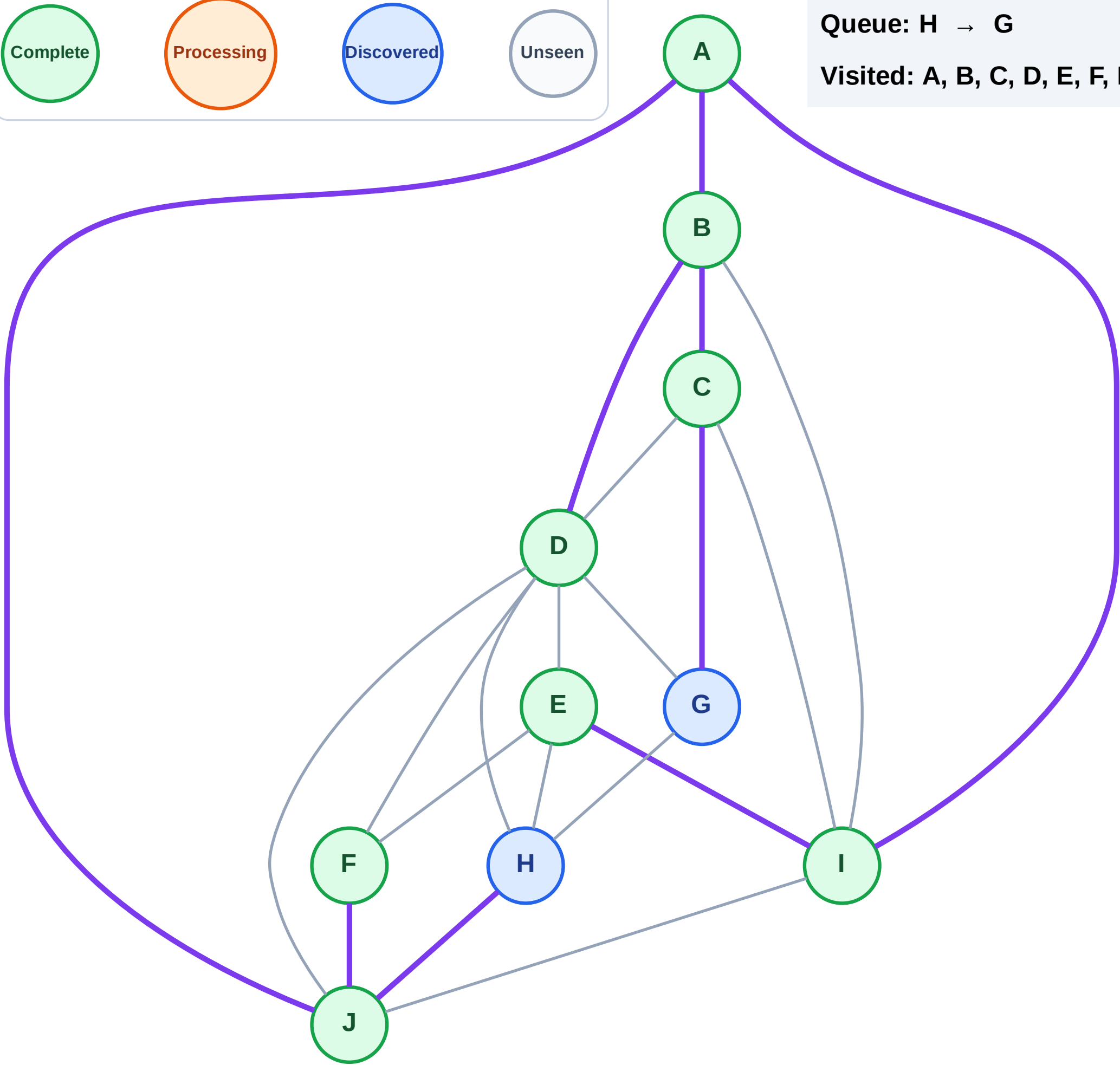
Step 17: No new neighbors from F

Legend



Queue: H → G

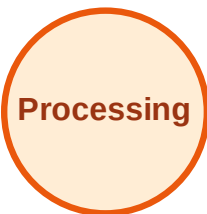
Visited: A, B, C, D, E, F, I, J



BFS Traversal

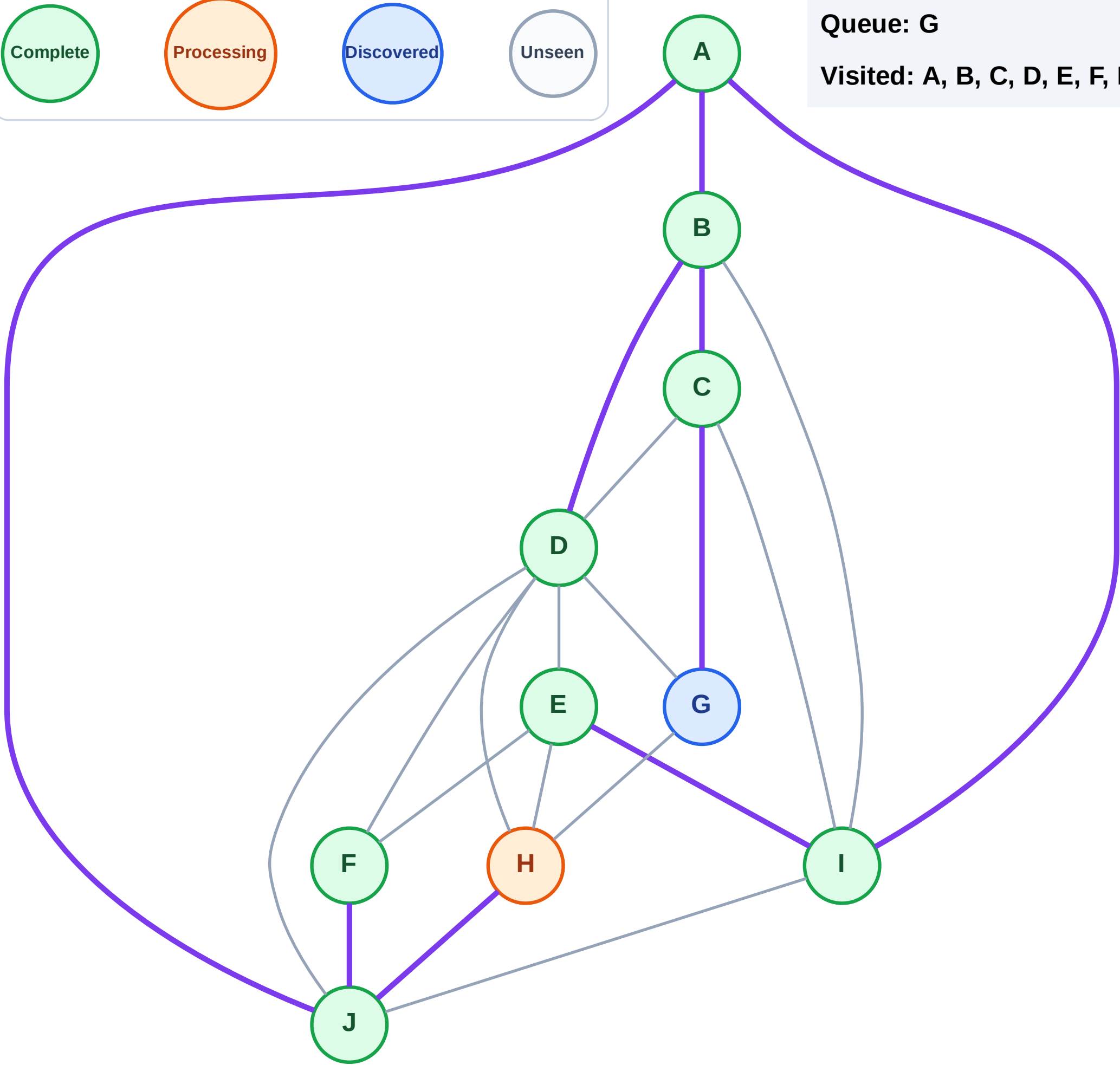
Step 18: Process node H

Legend



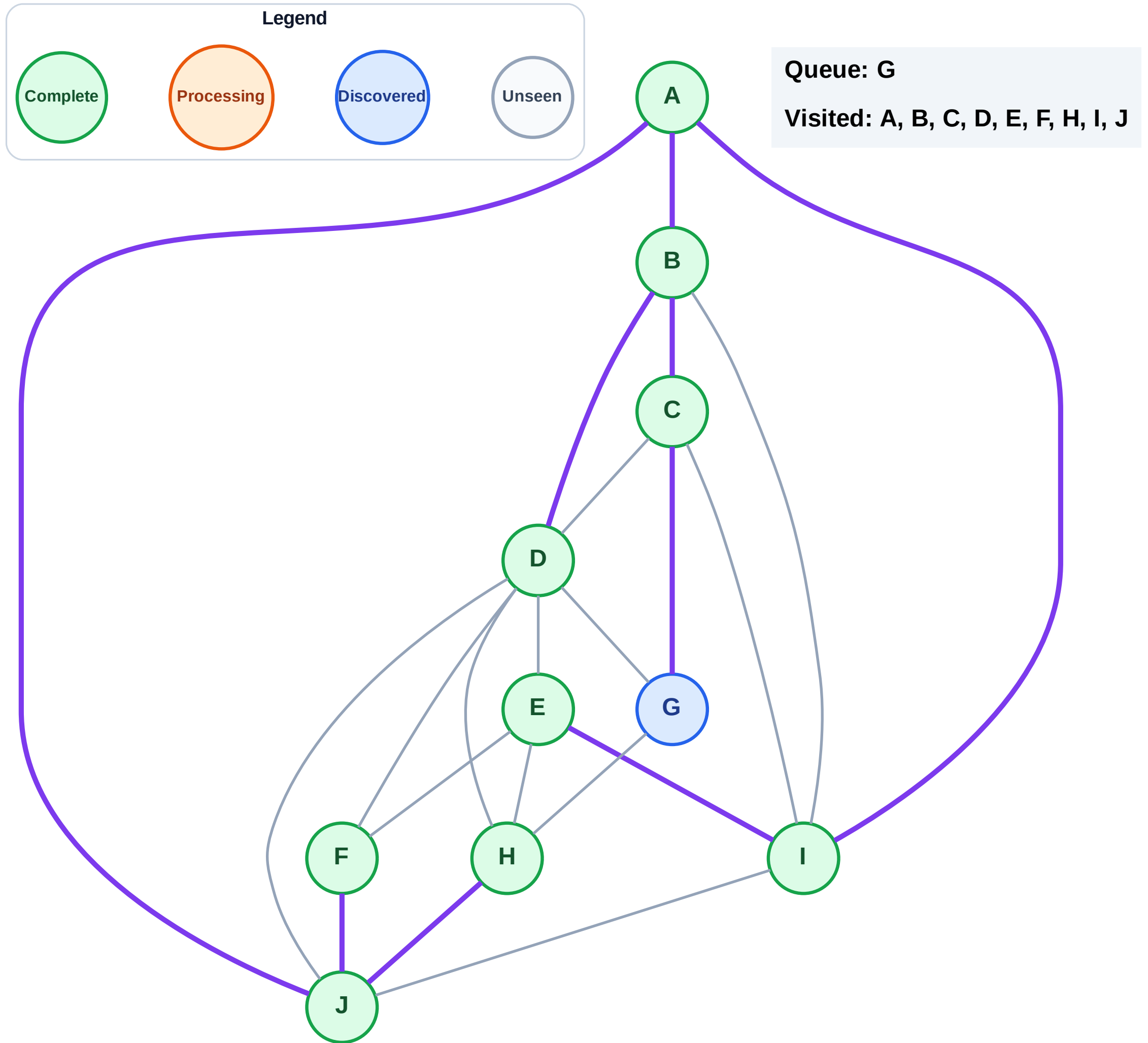
Queue: G

Visited: A, B, C, D, E, F, I, J



Step 19: No new neighbors from H

Step 19: No new neighbors from H



BFS Traversal

Step 20: Process node G

Legend

Complete

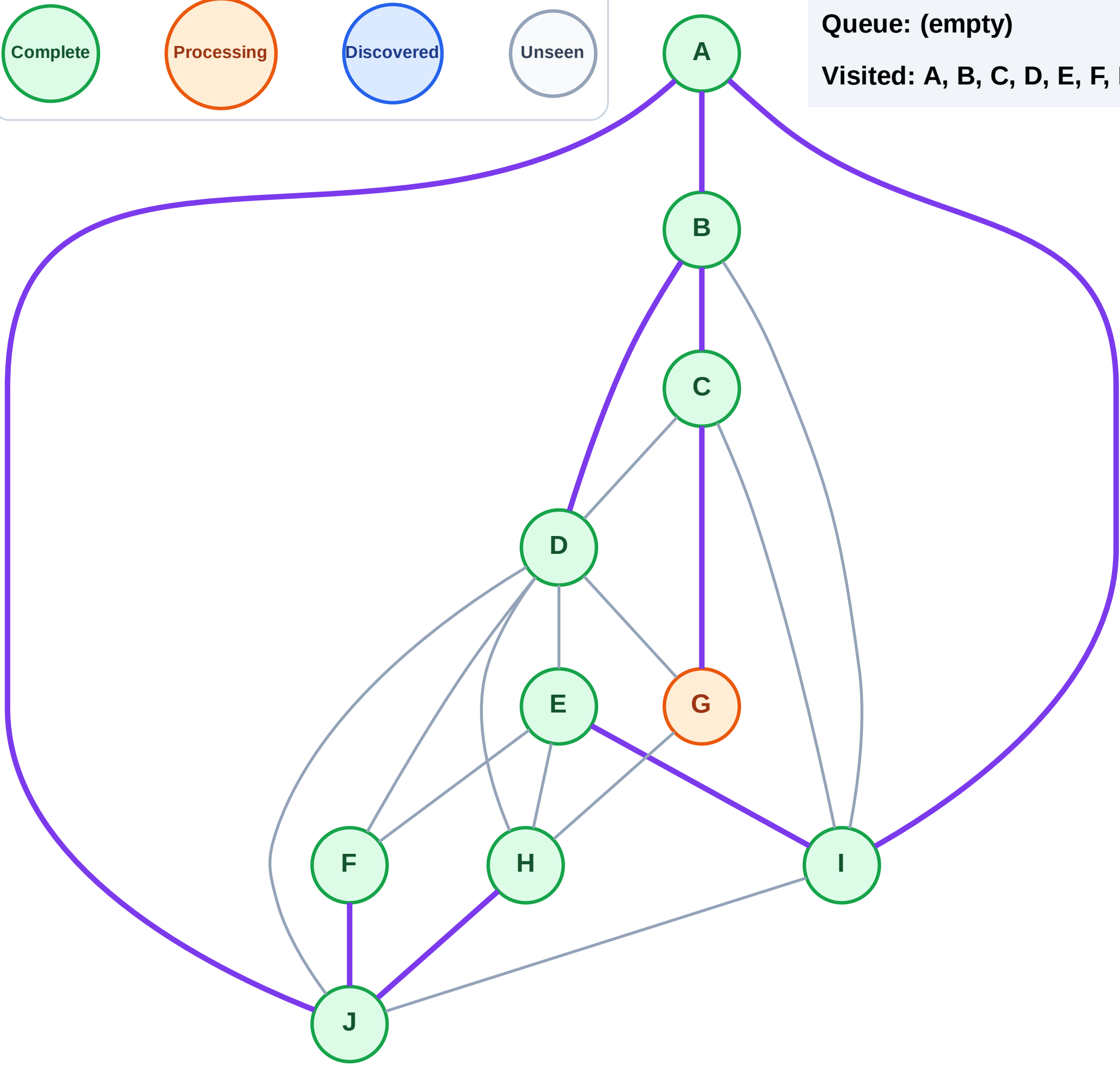
Processing

Discovered

Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

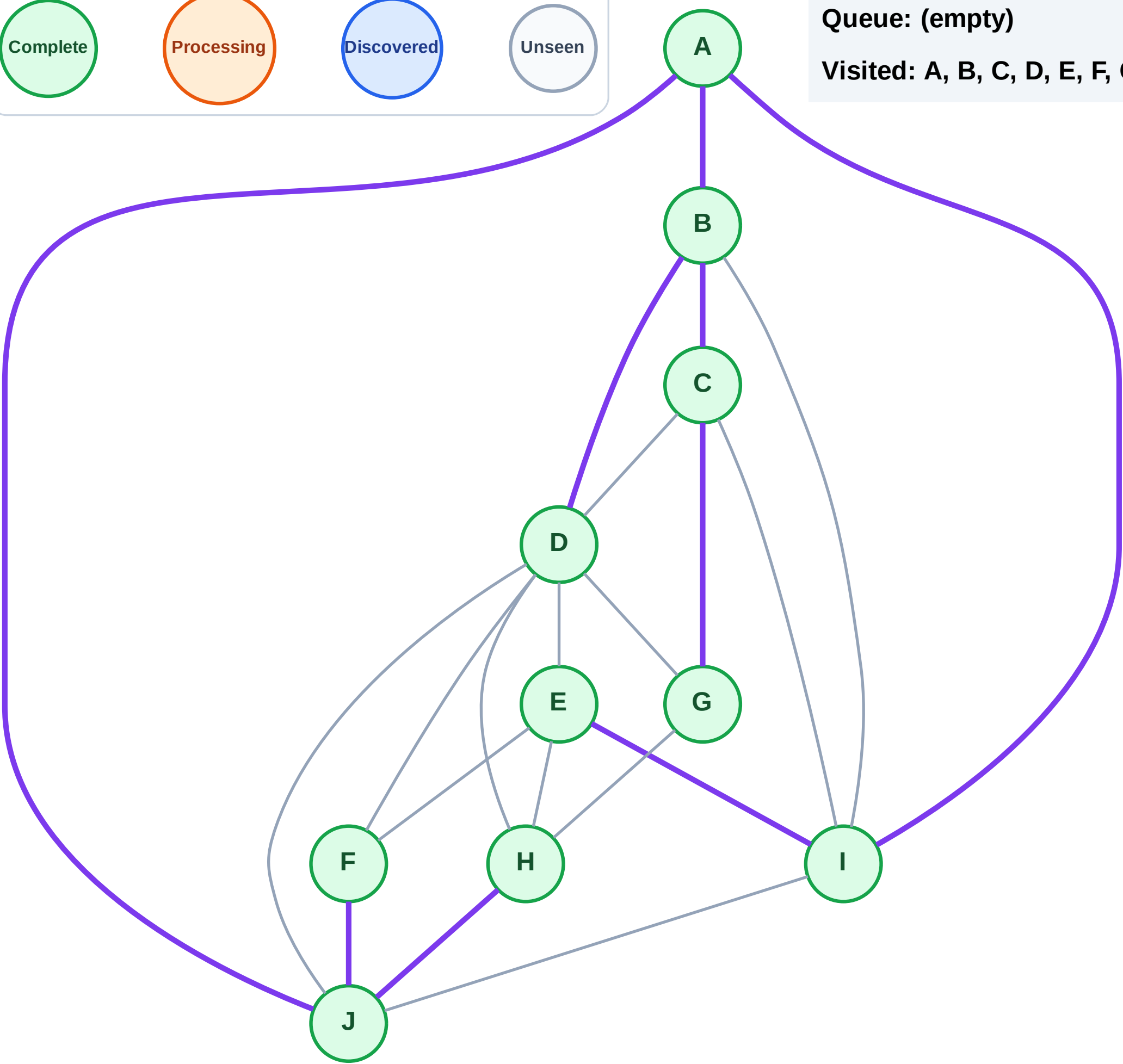
Step 21: No new neighbors from G

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

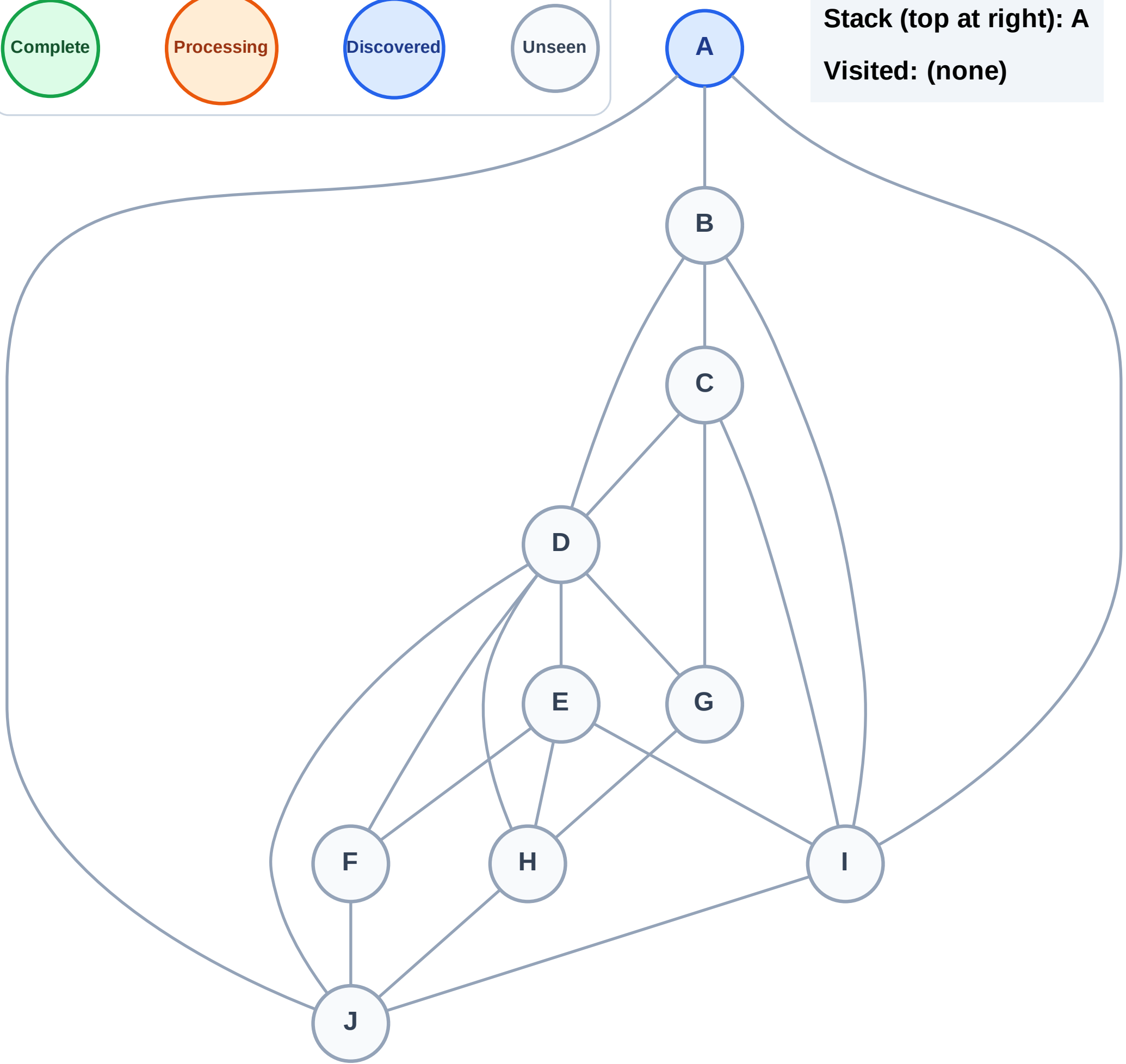
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

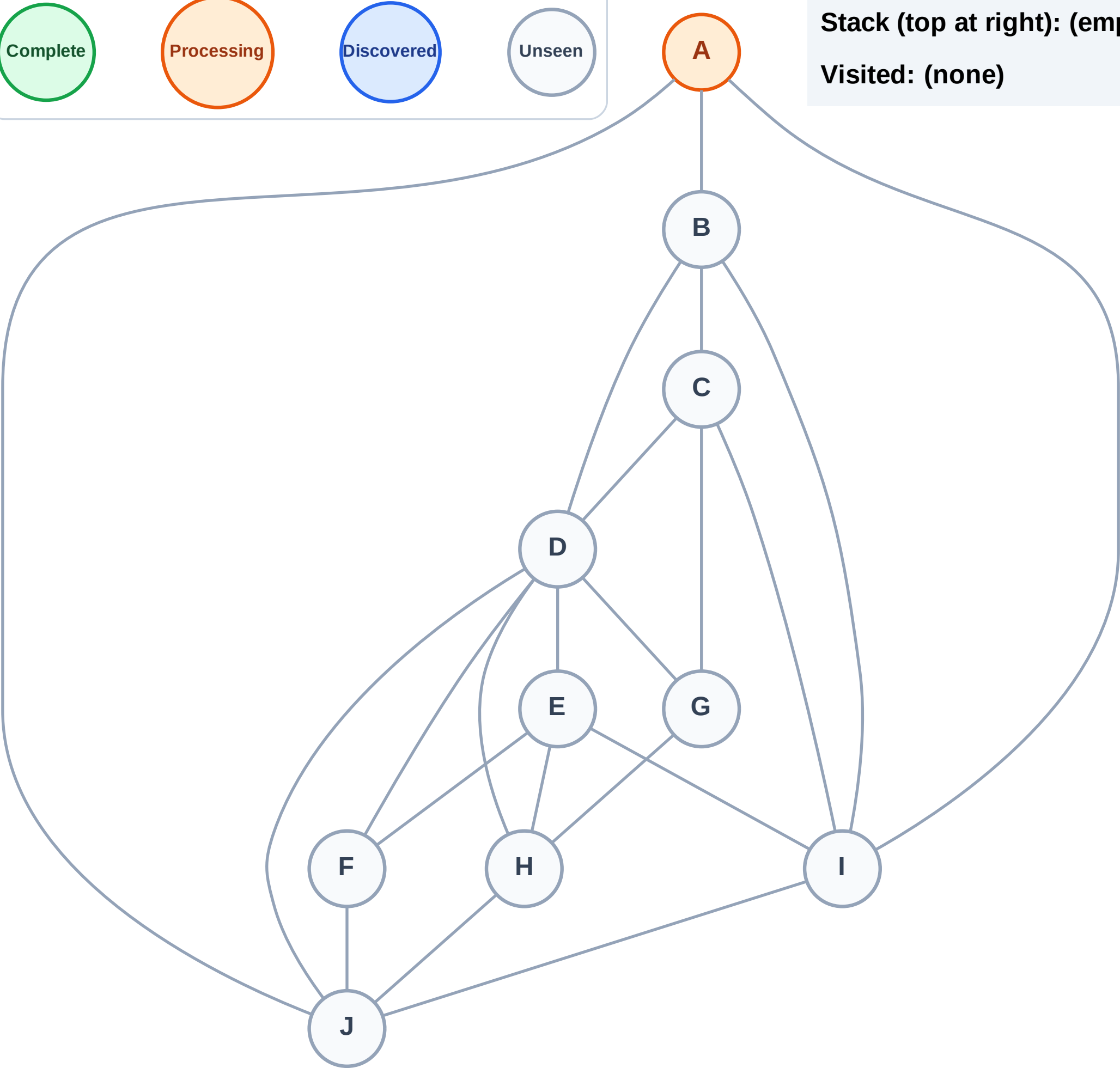
Complete

Processing

Discovered

Unseen

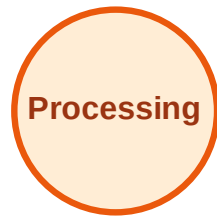
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

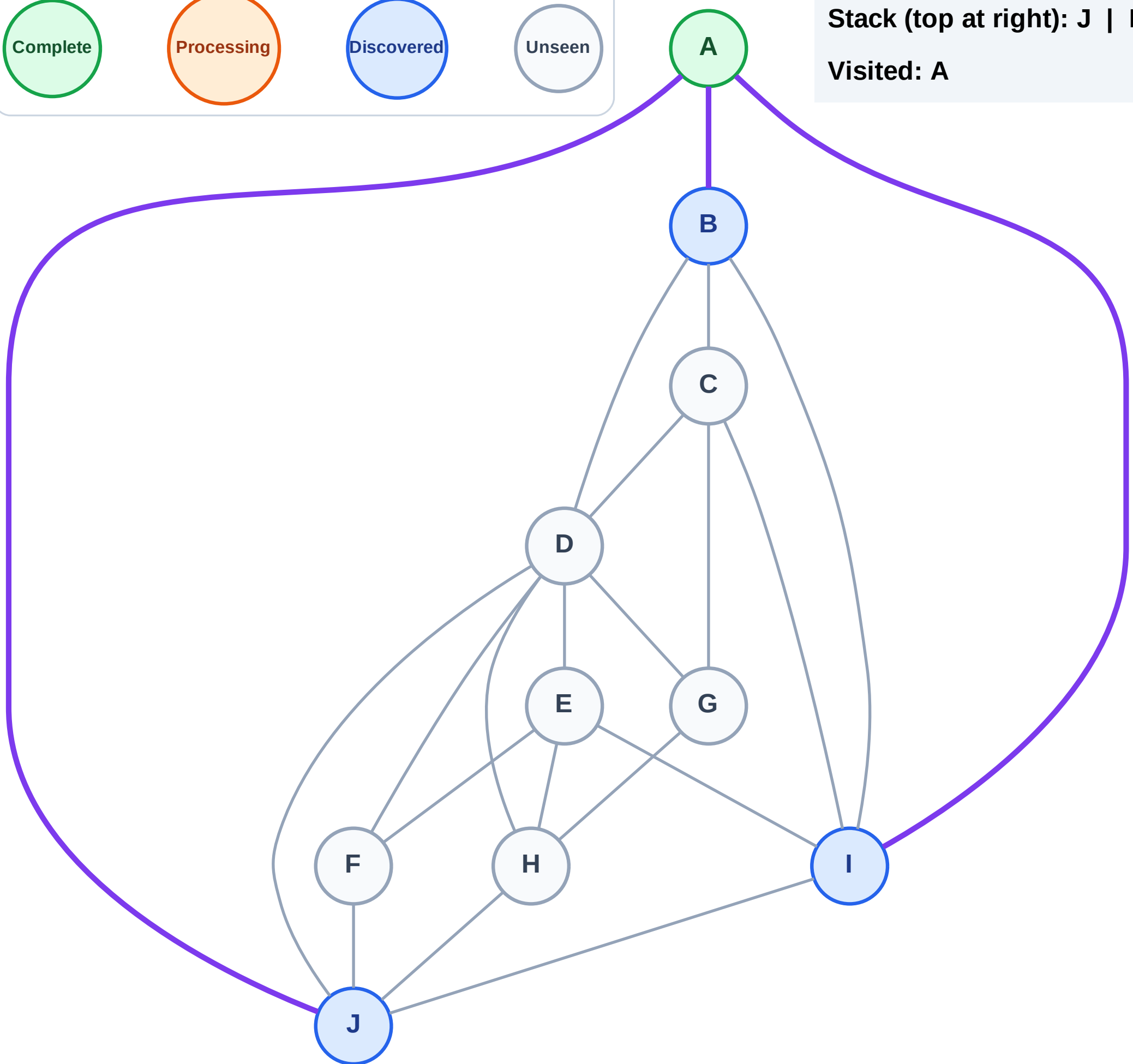
Step 3: Discover J, I, B from A

Legend



Stack (top at right): J | I | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

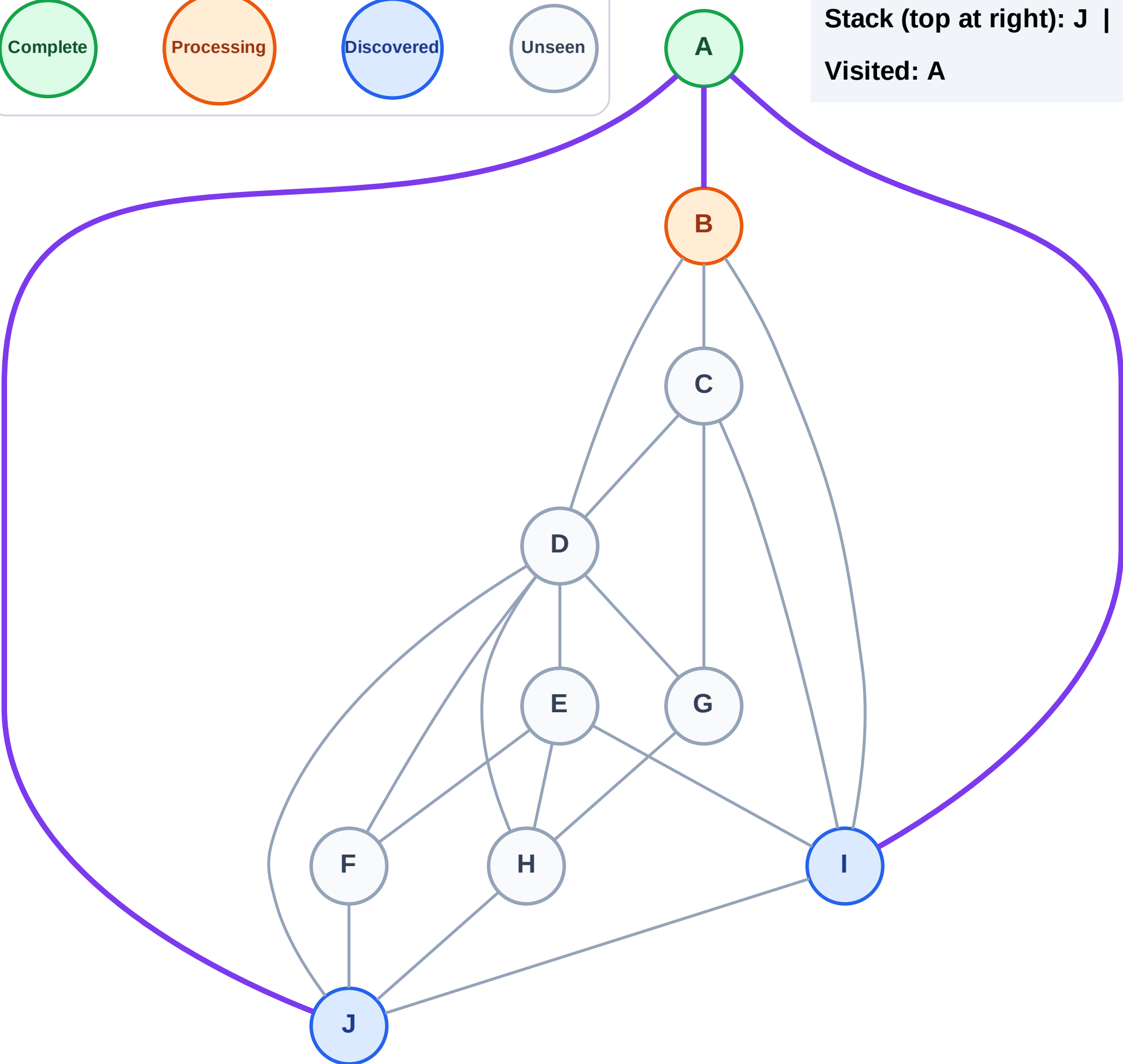
Processing

Discovered

Unseen

Stack (top at right): J | I

Visited: A



DFS Traversal

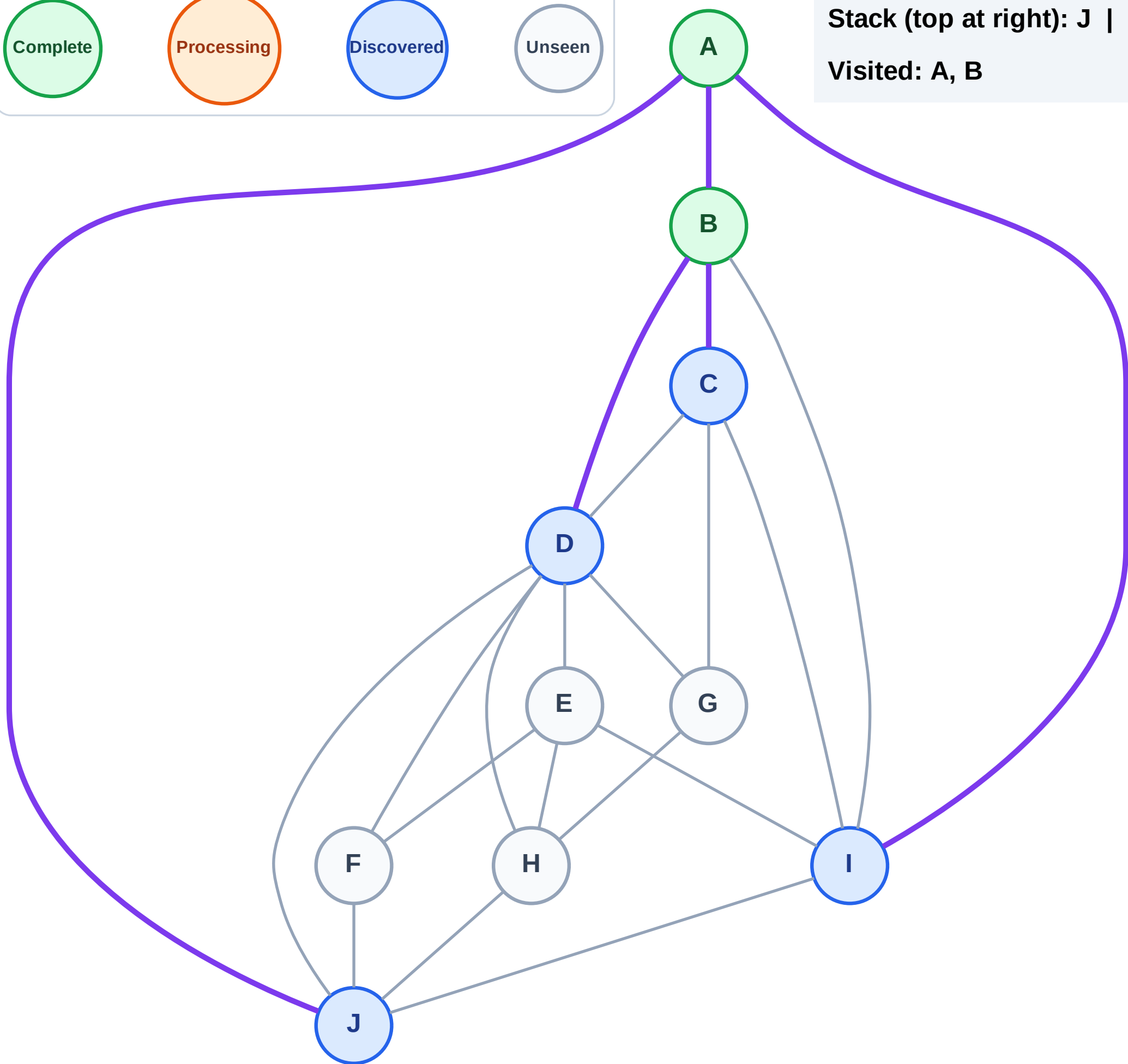
Step 5: Discover D, C from B

Legend



Stack (top at right): J | I | D | C

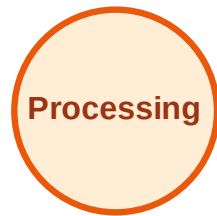
Visited: A, B



DFS Traversal

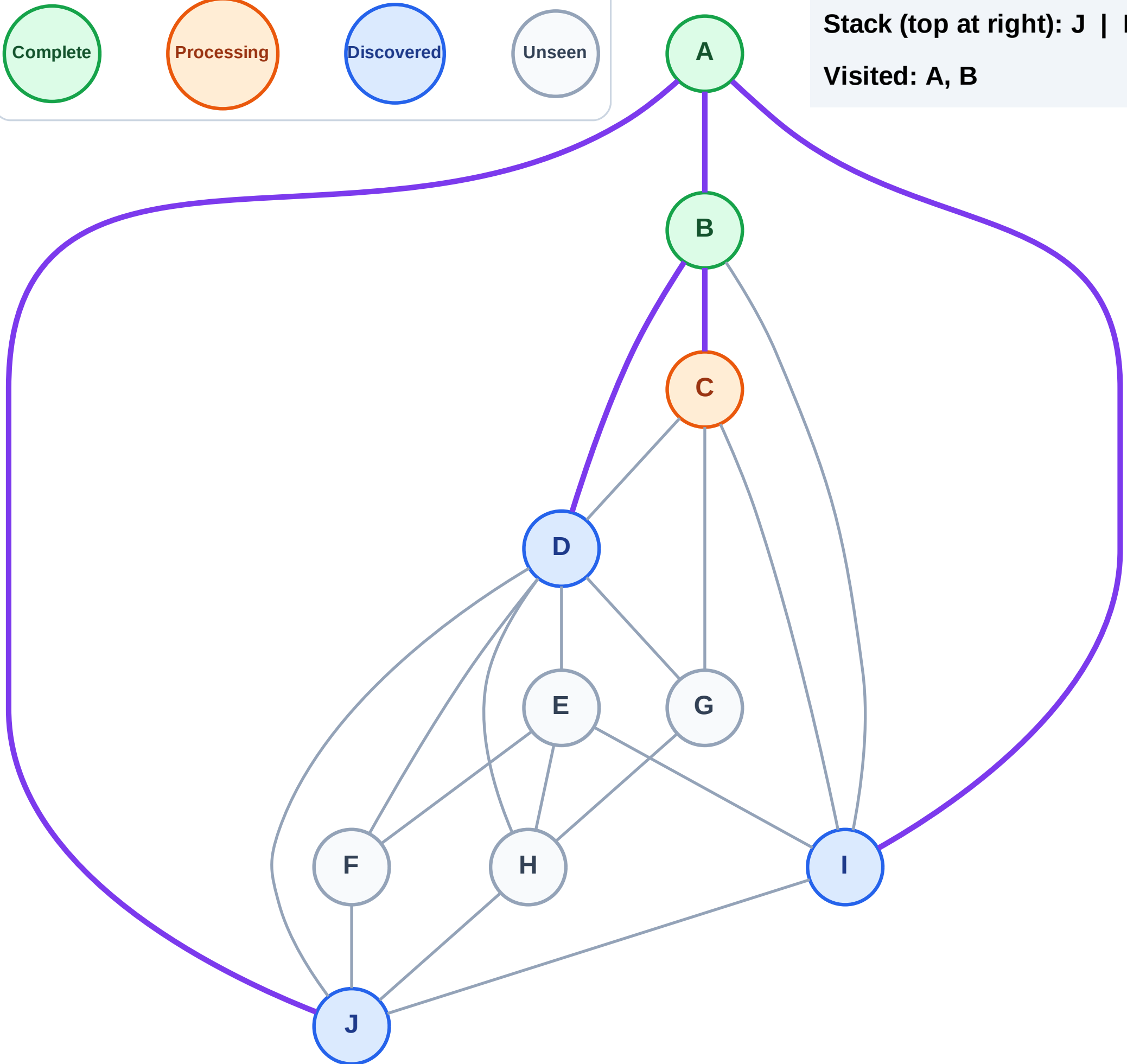
Step 6: Process node C

Legend



Stack (top at right): J | I | D

Visited: A, B



DFS Traversal

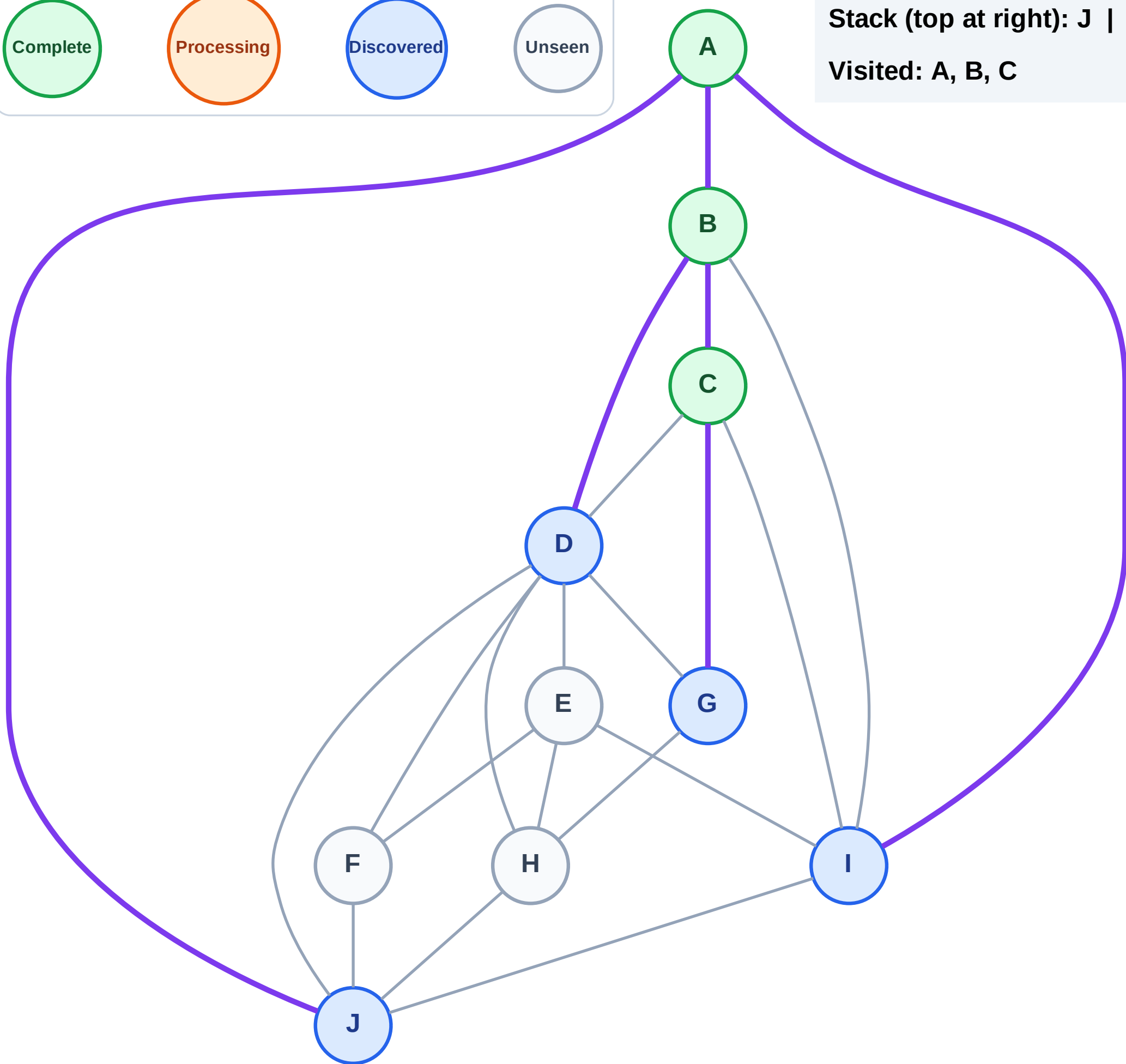
Step 7: Discover G from C

Legend



Stack (top at right): J | I | D | G

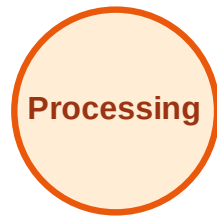
Visited: A, B, C



DFS Traversal

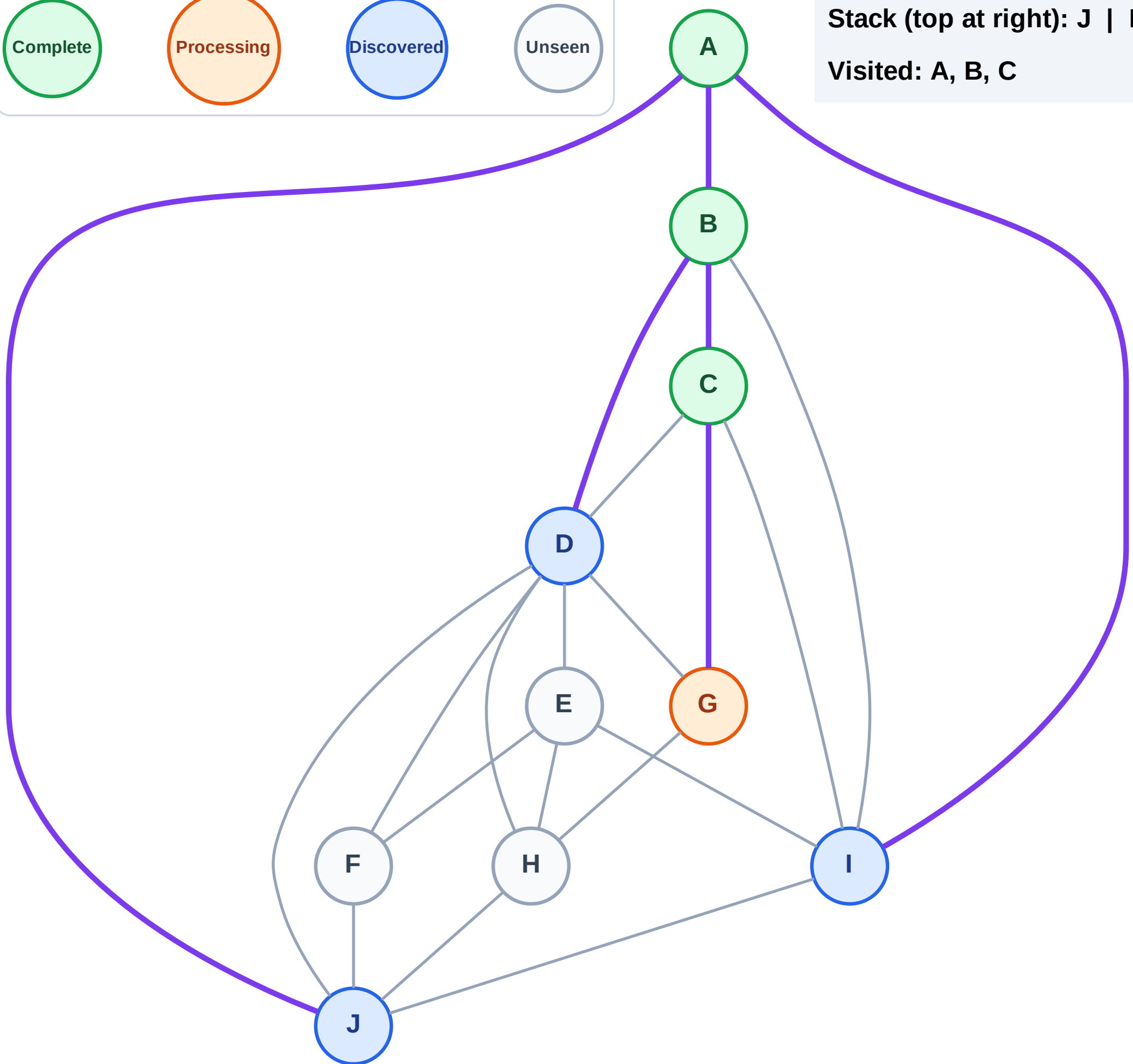
Step 8: Process node G

Legend



Stack (top at right): J | I | D

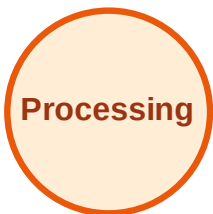
Visited: A, B, C



DFS Traversal

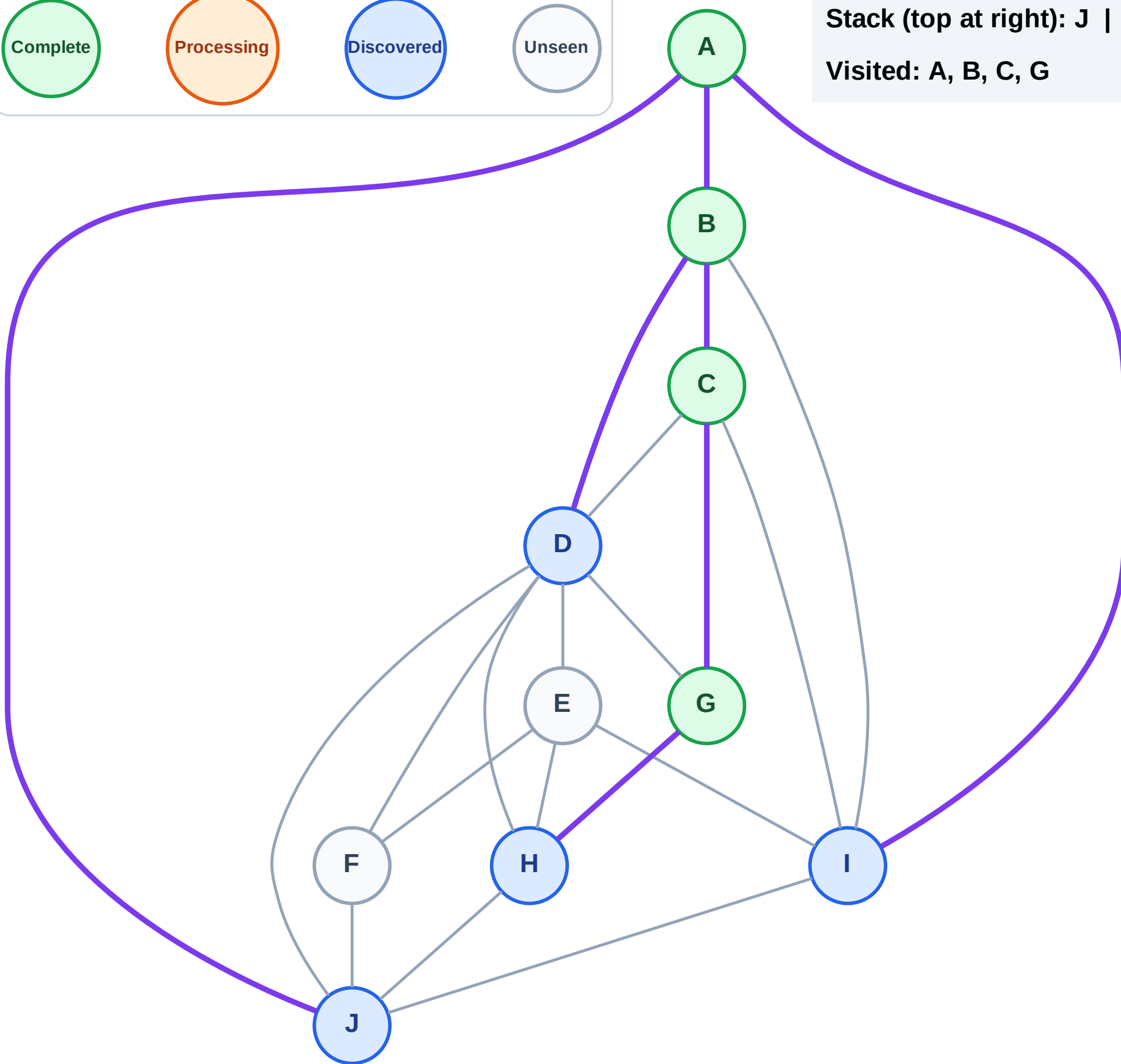
Step 9: Discover H from G

Legend



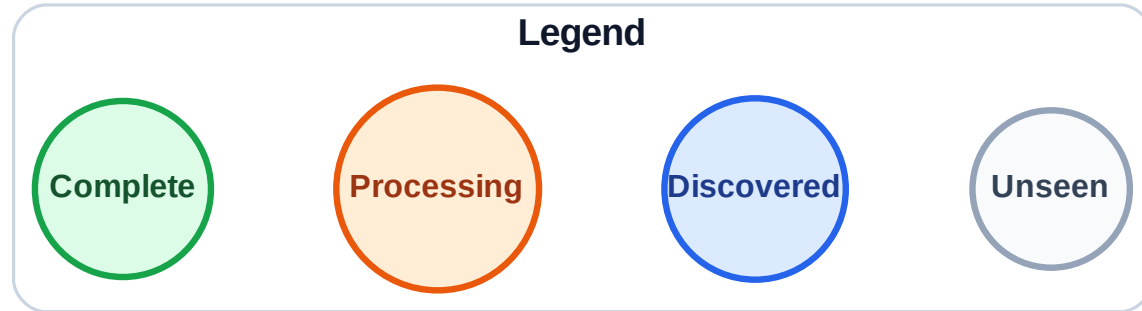
Stack (top at right): J | I | D | H

Visited: A, B, C, G



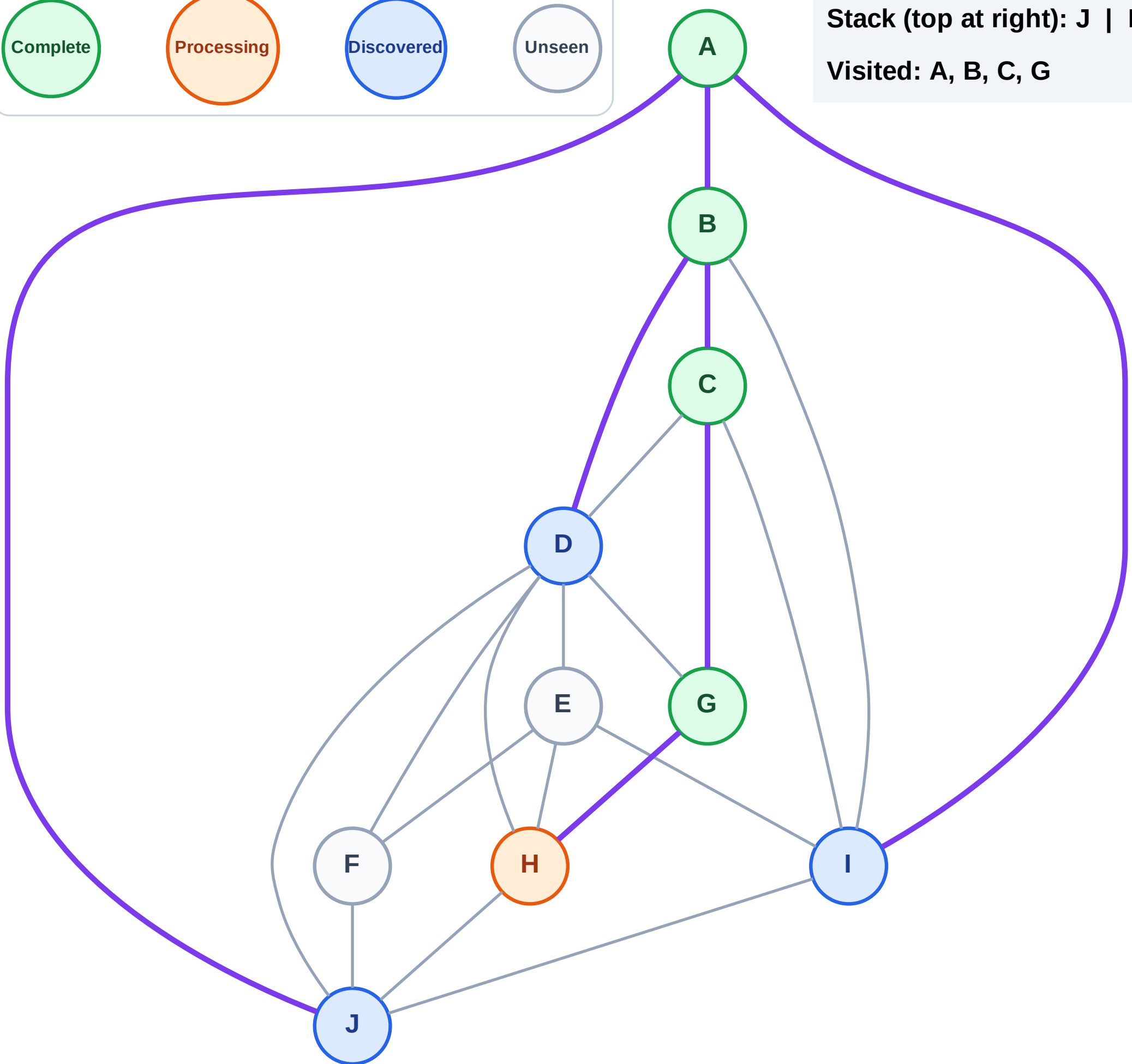
Step 10: Process node H

Step 10: Process node H



Stack (top at right): J | I | D

Visited: A, B, C, G



DFS Traversal

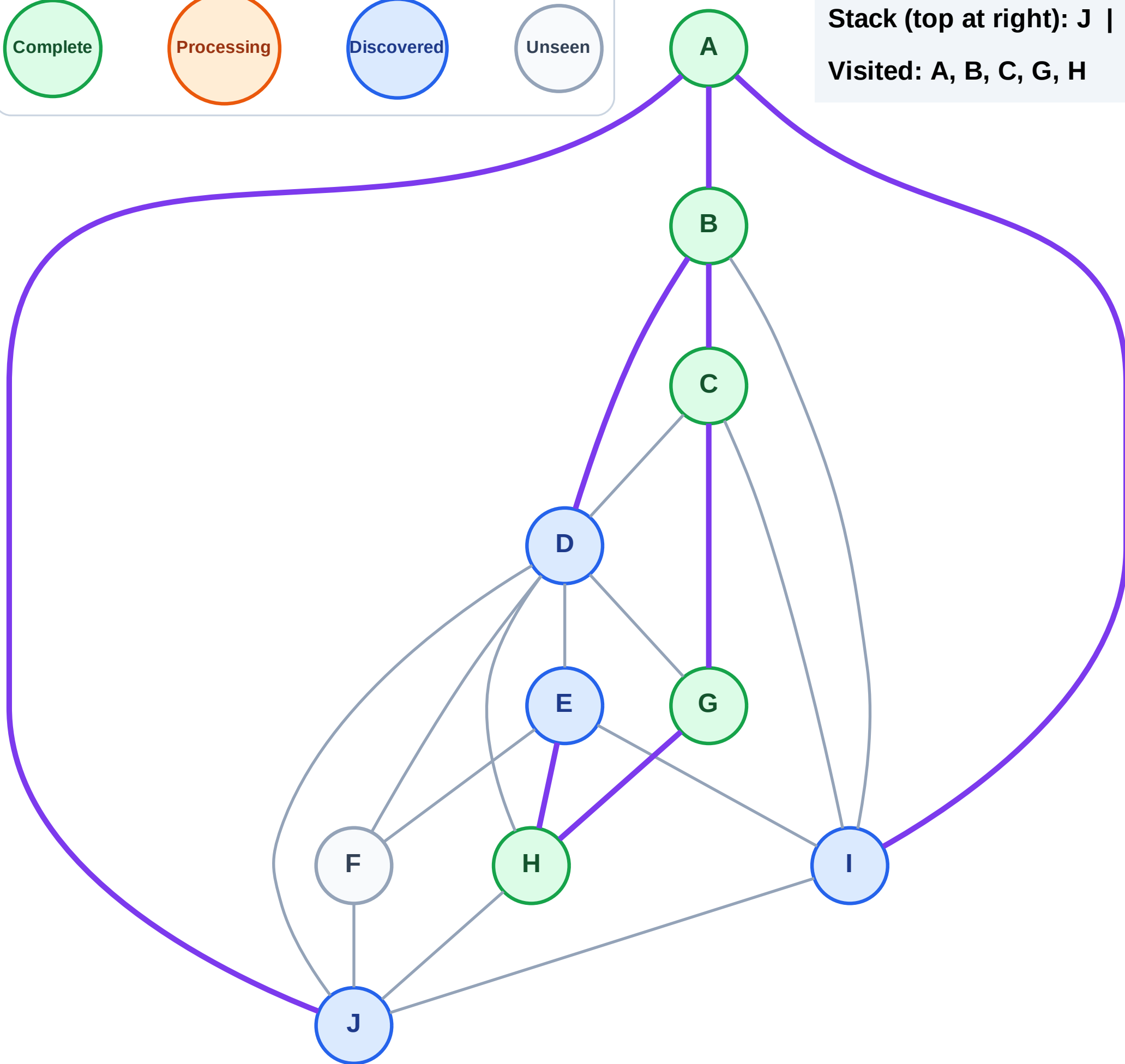
Step 11: Discover E from H

Legend



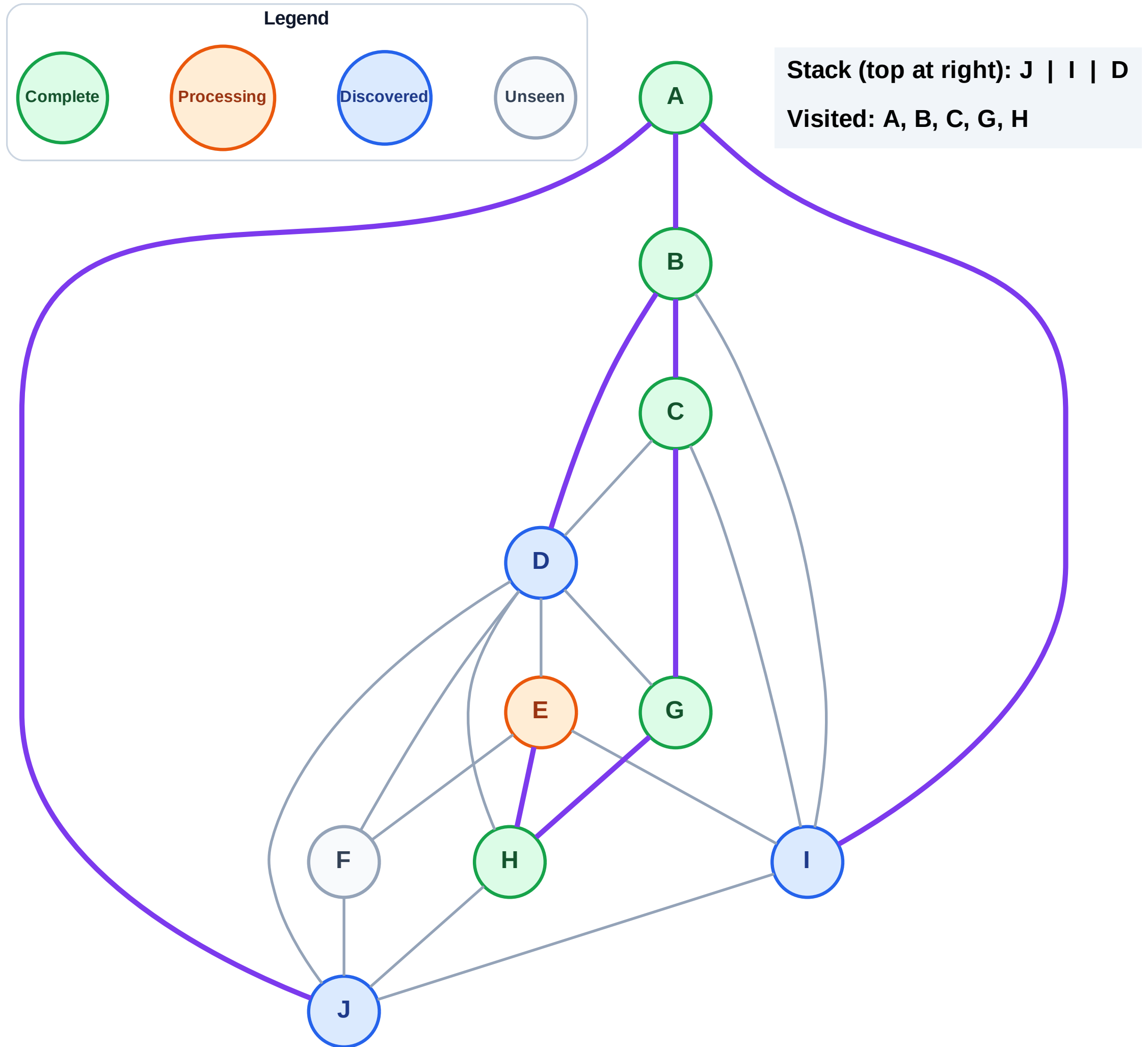
Stack (top at right): J | I | D | E

Visited: A, B, C, G, H



Step 12: Process node E

Step 12: Process node E



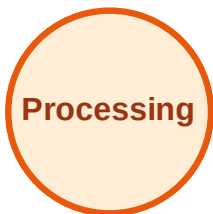
DFS Traversal

Step 13: Discover F from E

Legend



Complete



Processing



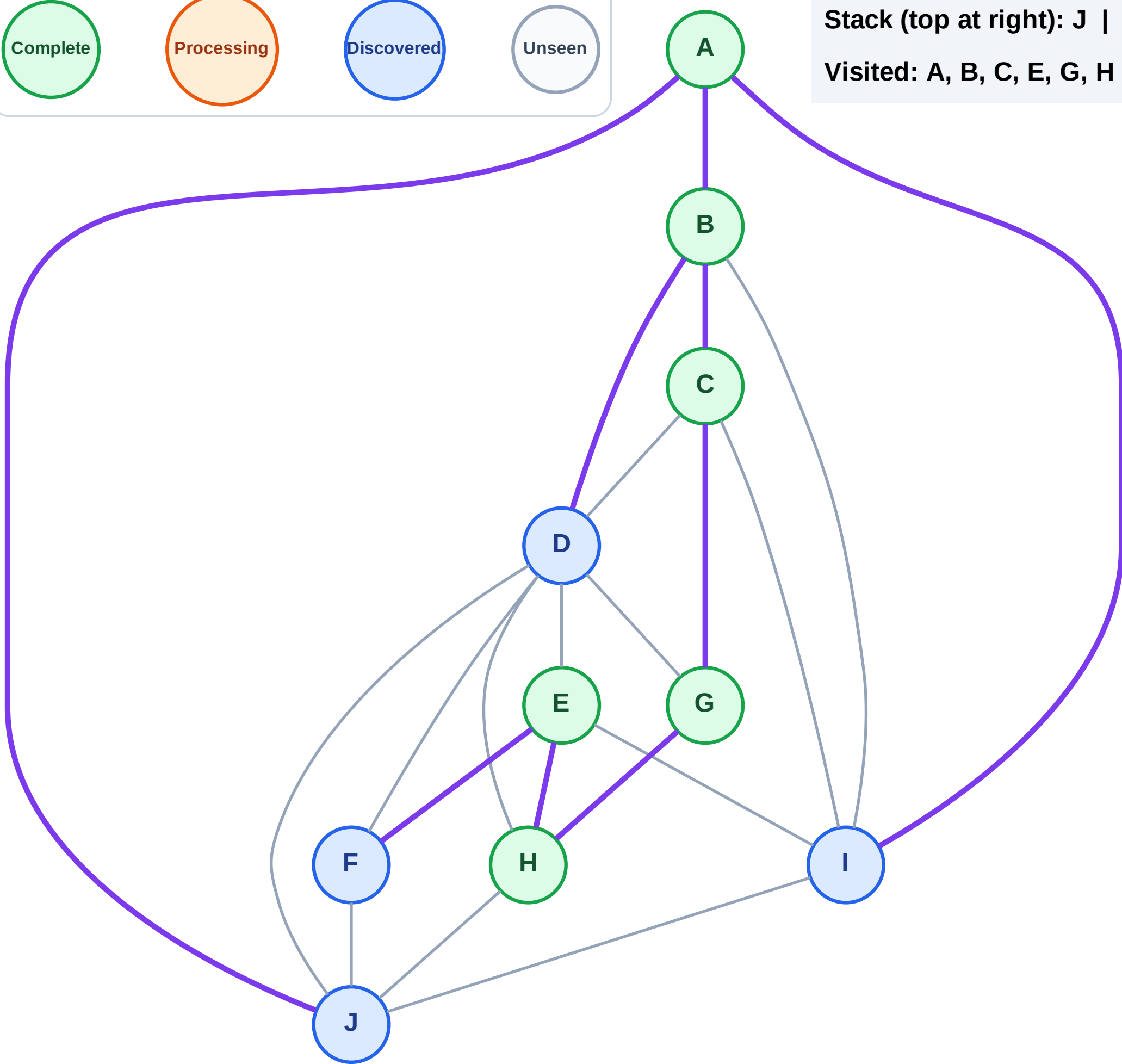
Discovered



Unseen

Stack (top at right): J | I | D | F

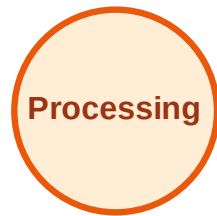
Visited: A, B, C, E, G, H



DFS Traversal

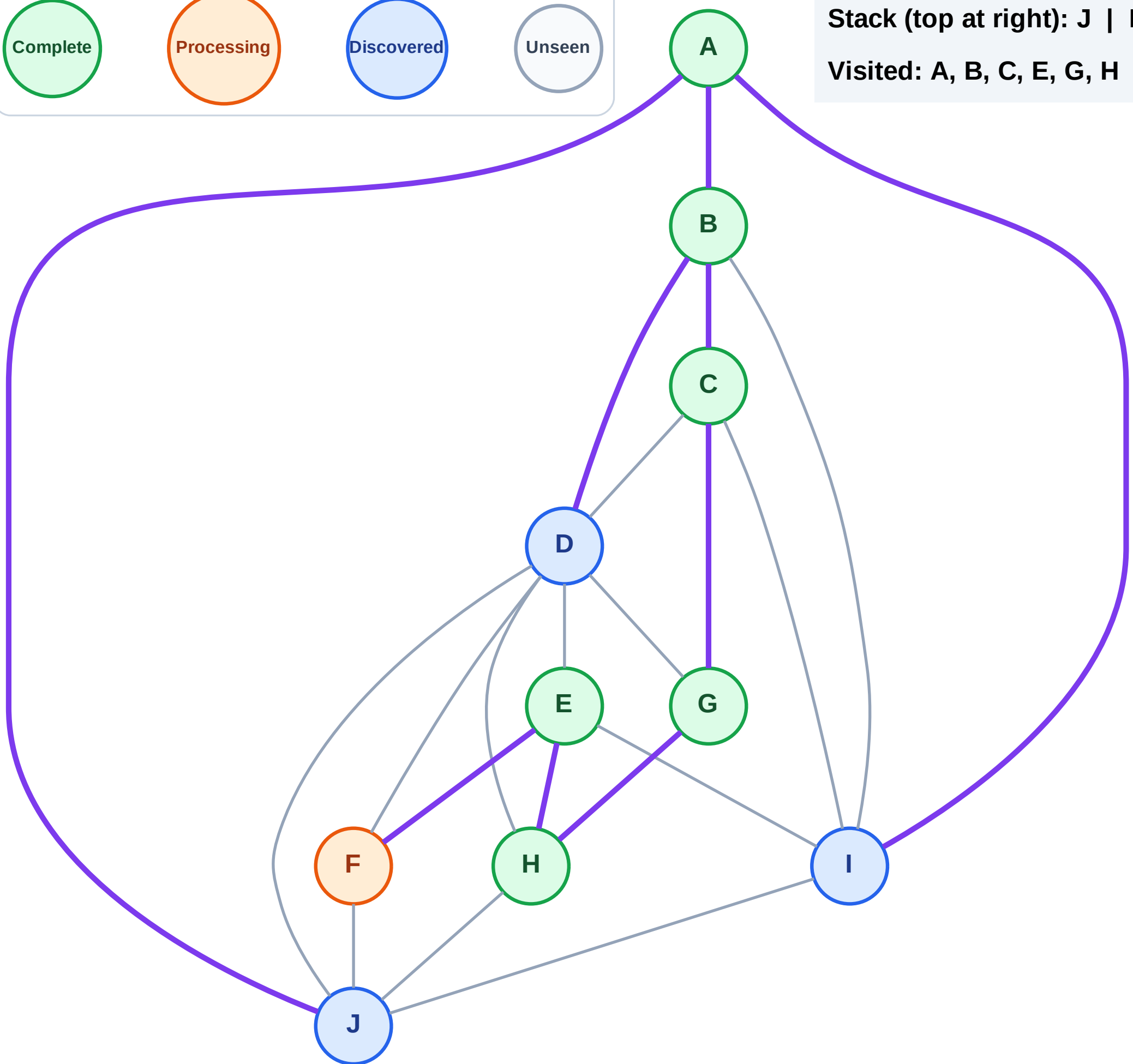
Step 14: Process node F

Legend



Stack (top at right): J | I | D

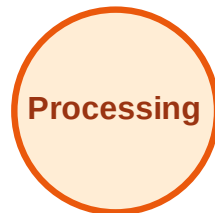
Visited: A, B, C, E, G, H



DFS Traversal

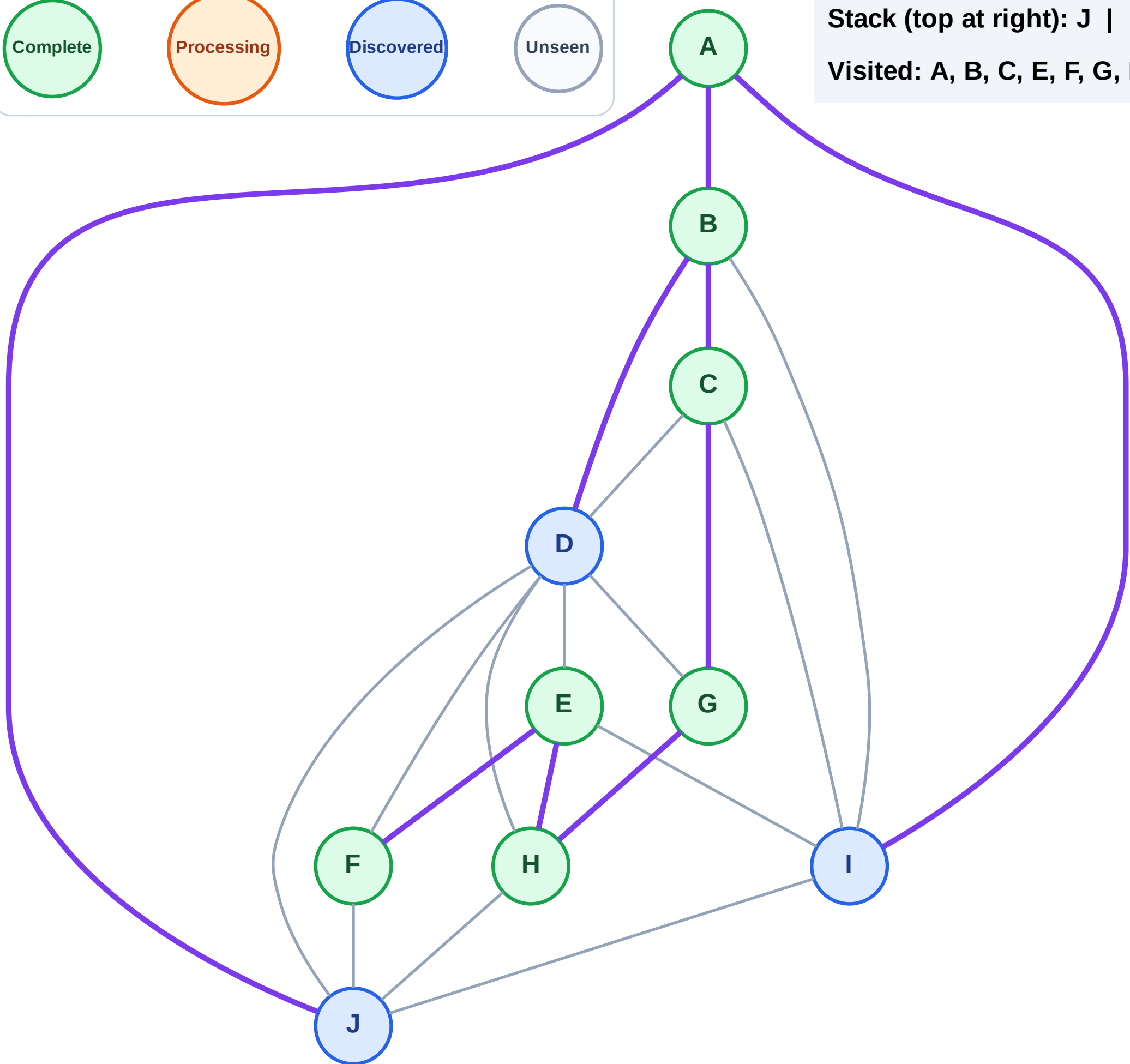
Step 15: No new neighbors from F

Legend



Stack (top at right): J | I | D

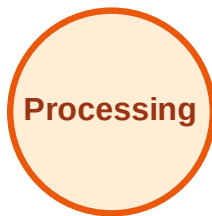
Visited: A, B, C, E, F, G, H



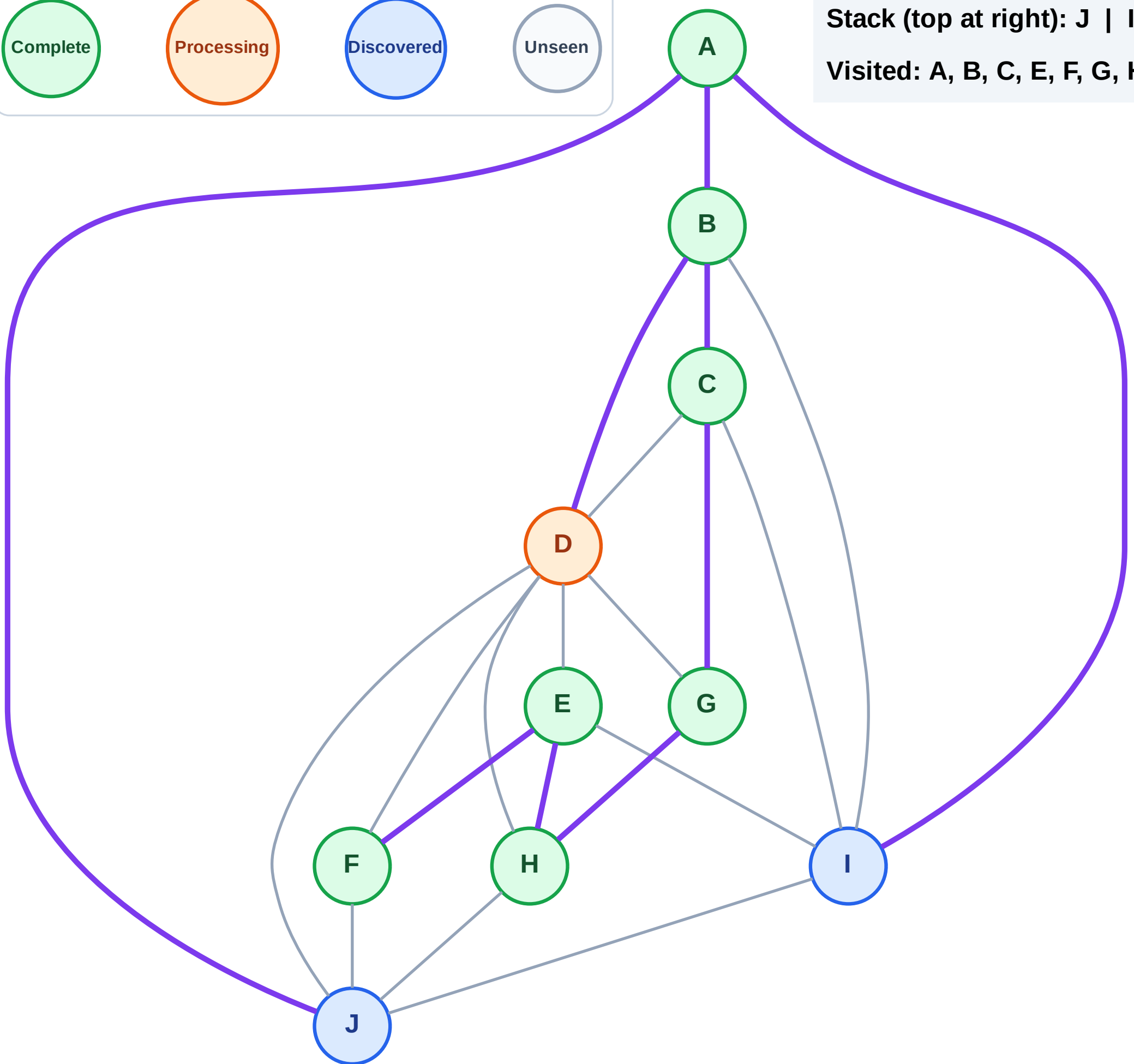
DFS Traversal

Step 16: Process node D

Legend



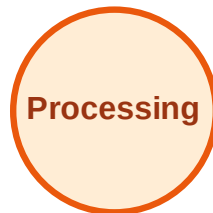
Stack (top at right): J | I
Visited: A, B, C, E, F, G, H



DFS Traversal

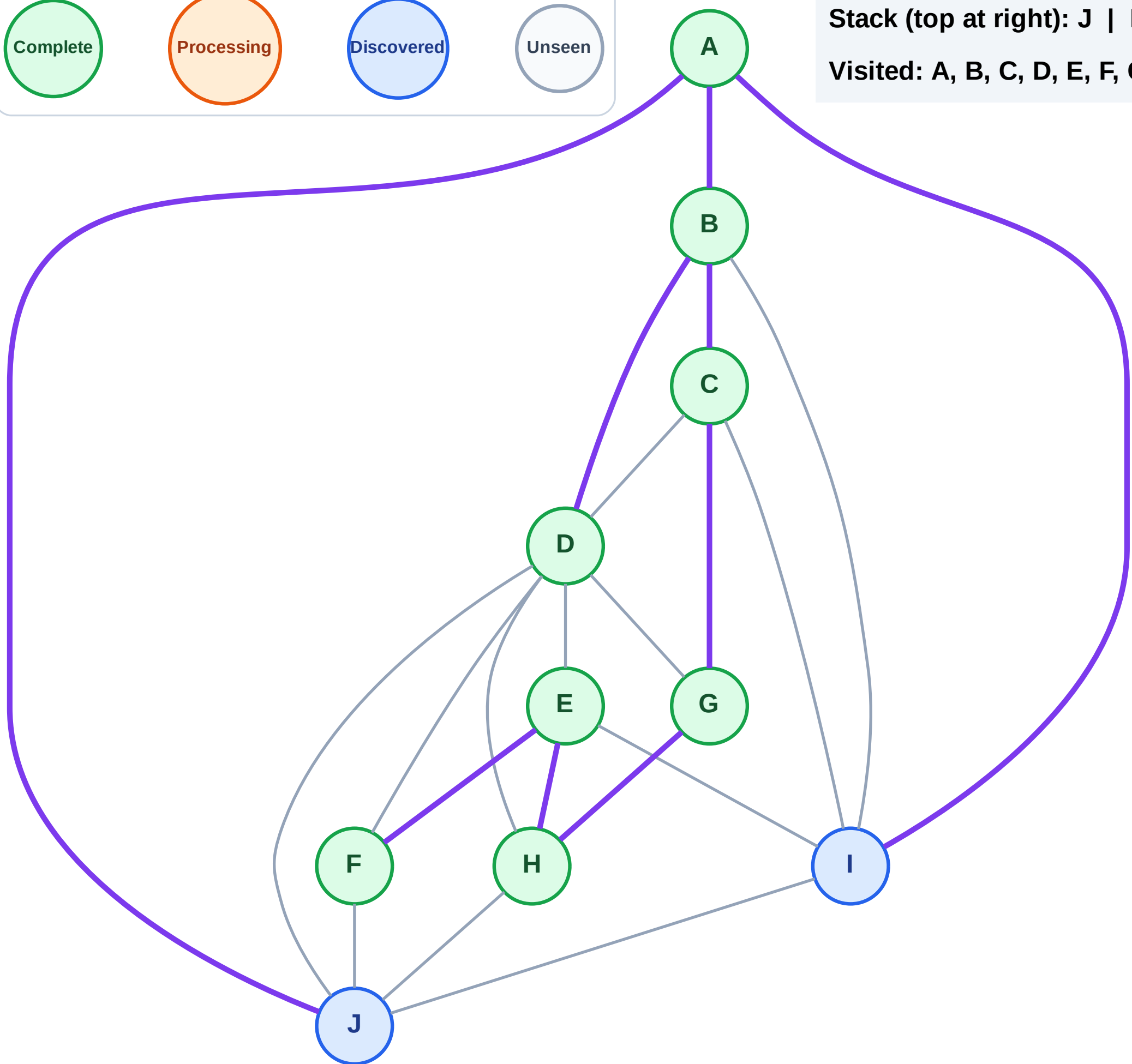
Step 17: No new neighbors from D

Legend



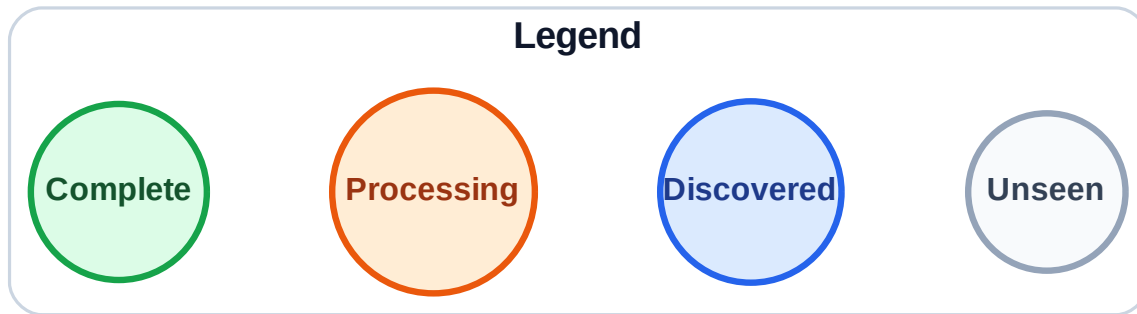
Stack (top at right): J | I

Visited: A, B, C, D, E, F, G, H



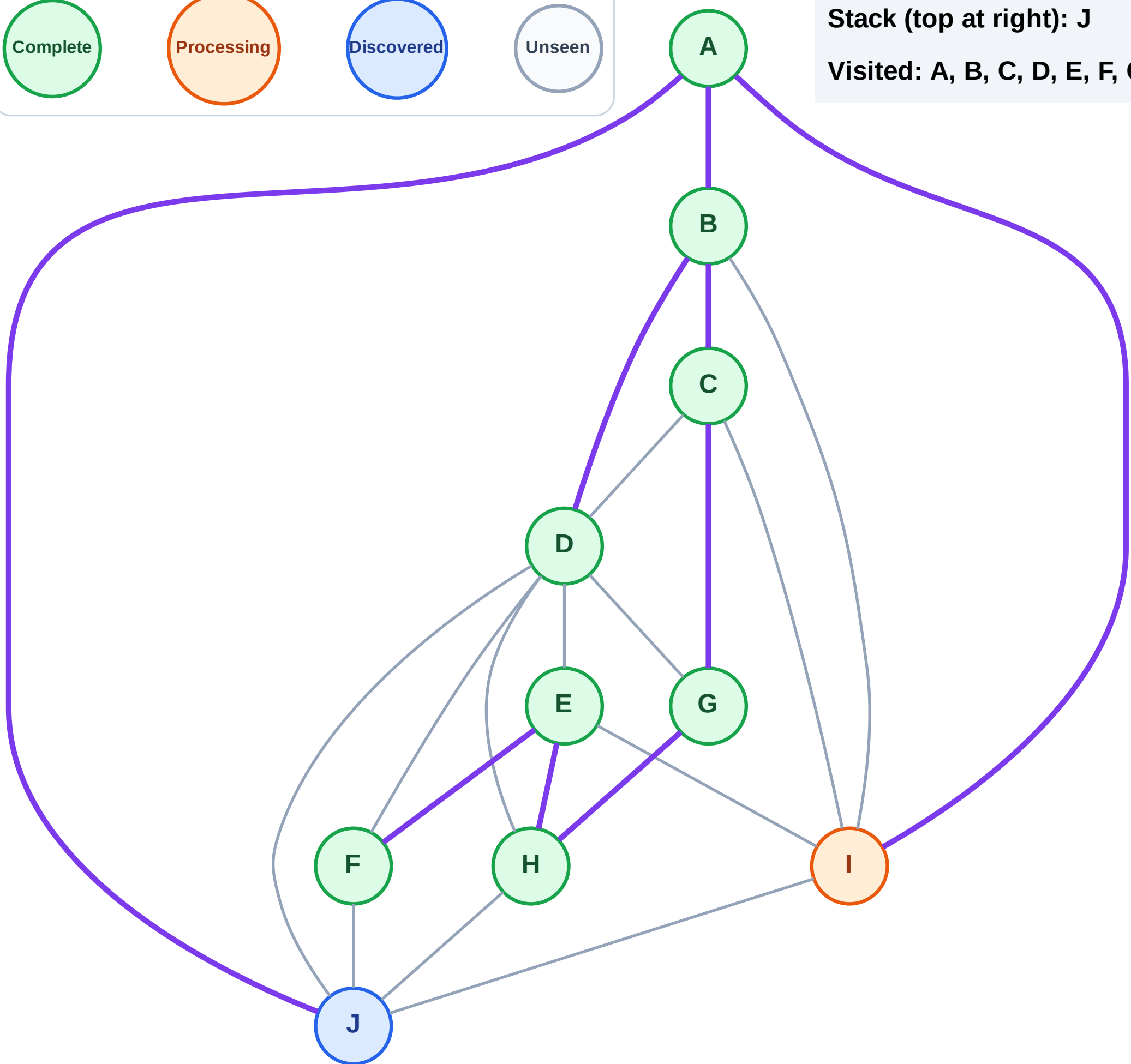
Step 18: Process node I

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Stack (top at right): J

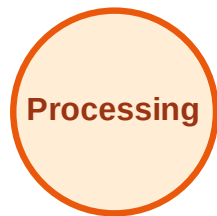
Visited: A, B, C, D, E, F, G, H



DFS Traversal

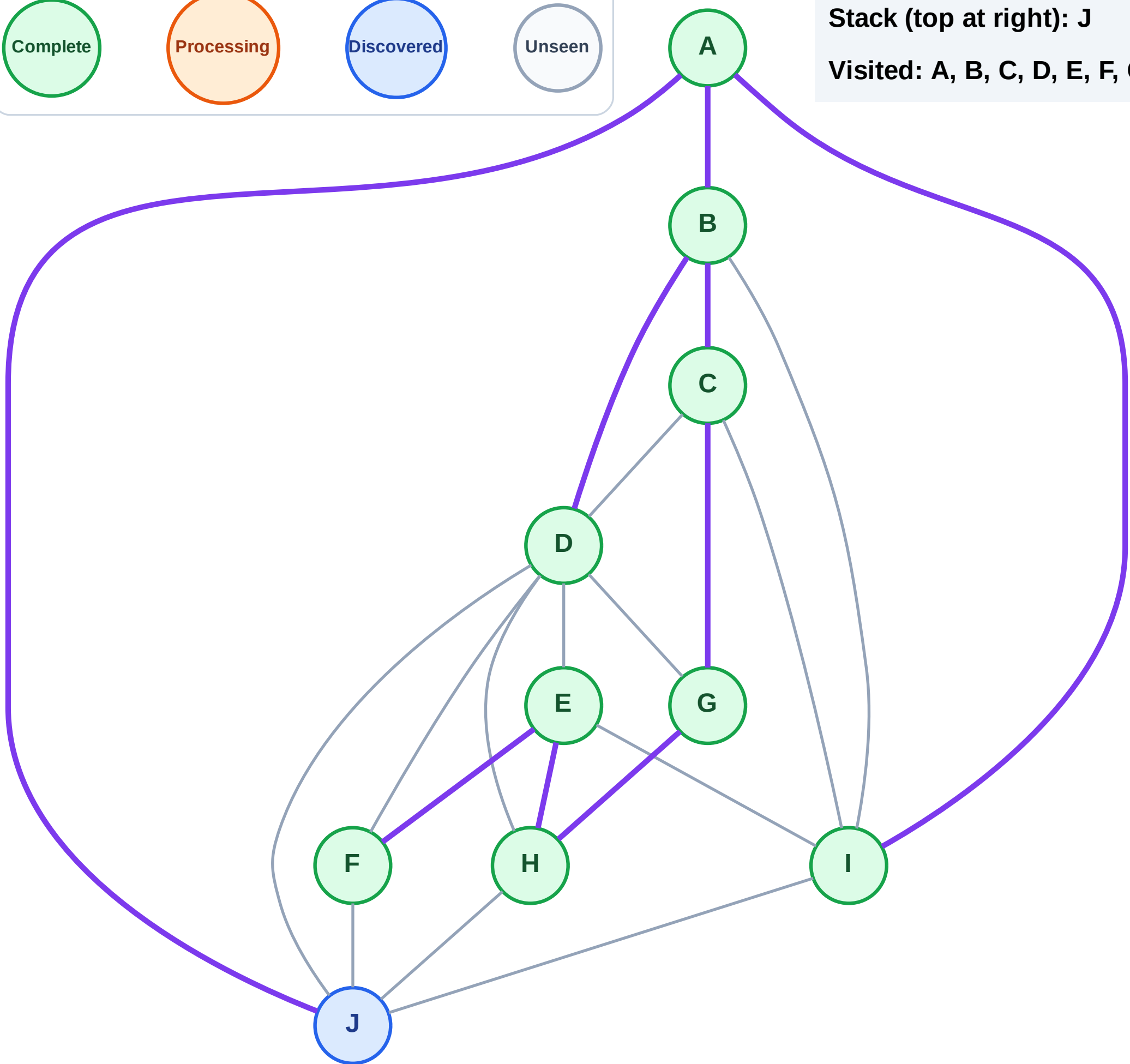
Step 19: No new neighbors from I

Legend



Stack (top at right): J

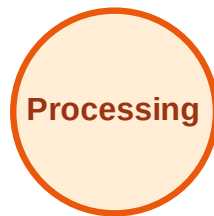
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

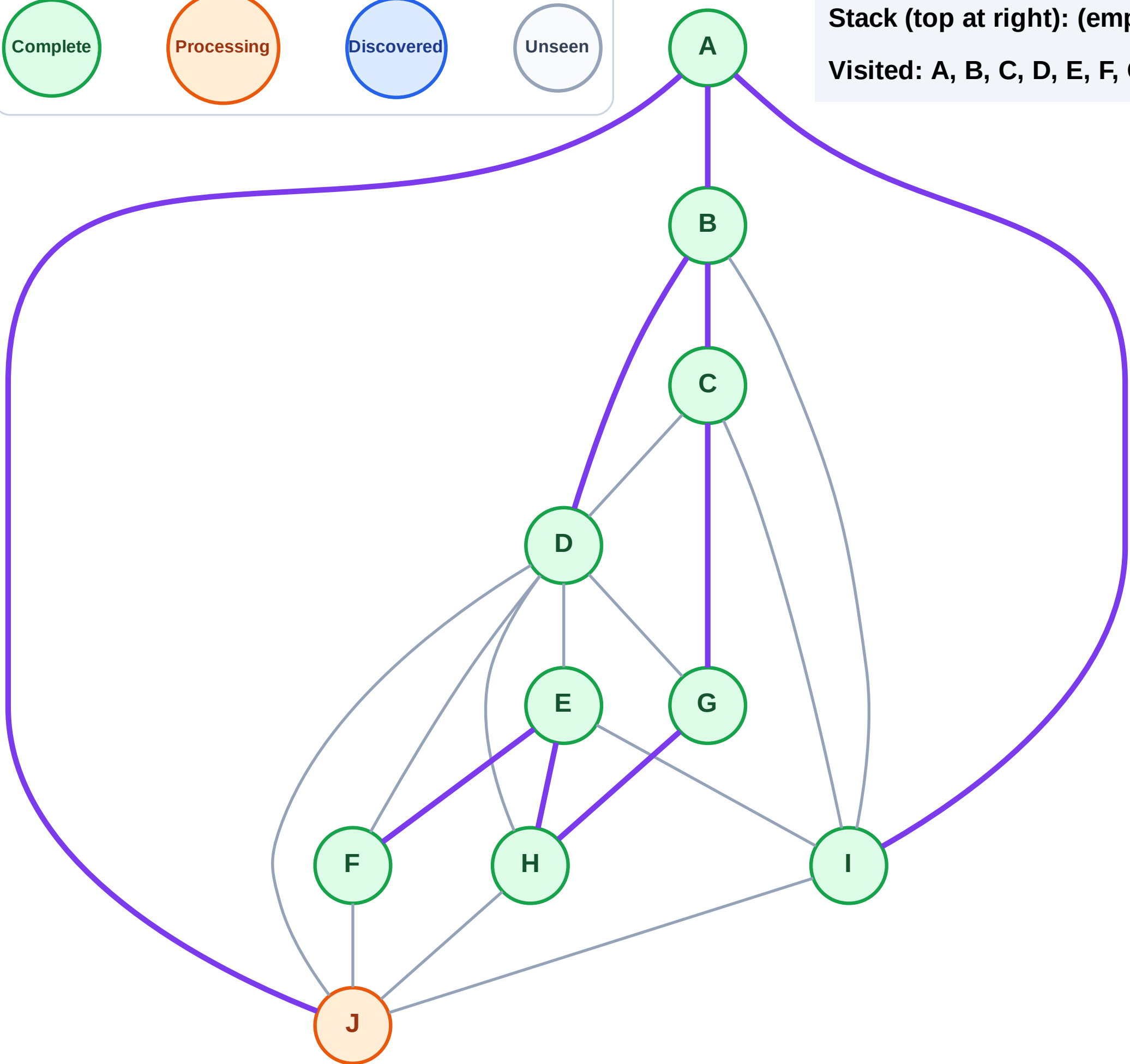
Step 20: Process node J

Legend



Stack (top at right): (empty)

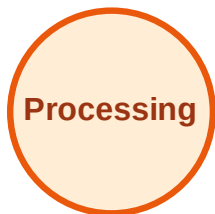
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

